

The Rank-Seven Parent Packet: Support–Port Reconstruction, Tau-Local Composition, and Class-Minimal Realization

Tau Core Technical Paper VII-B

Jozsef Olcsak

Last revised: 18 September 2026

Abstract

This technical paper isolates the algebraic and source-law content previously embedded in the joint gravity–matter–gauge manuscript. Starting from a finite typed support with two ordered cotangent ports, one common unit and a noncollapsing body decoder, we reconstruct a seven-dimensional physical quotient. A pointed exterior-seed realization supplies the canonical cyclic record carrier, while a faithful nonracial control proves that rank and central weights alone do not fix its metric. An orthogonally invariant, positive operator-square functional has the Clifford relation as its exact zero set. We also prove a Tau-local subsystem-composition theorem with hidden global-relation and crossing-rule controls, and construct a class-minimal enriched law whose global minimum occupies the complete packet inside the declared finite source-irredundant one-copy class. The unchanged narrow reduct does not entail this law, and no theorem here proves unrestricted physical base–seed realization, ambient-parent exhaustivity, Standard-Model occupation, or a measured Tau-specific anomaly. A frozen public-source audit finds no currently executable gain-free cross-terminal endpoint.

Inherited morphology terminology. The upstream body is Paper I’s complete *stabilized morphological response configuration*, selected from a base carrier and a base-readable seed/load pattern. “Morphological body” is shorthand. The rank-seven source packet does not identify the body with the base and supplies neither a literal morphogenetic trajectory nor observer time. Under MRC-DEF2, these parent-side objects are typed pregeometric relational candidates, not ordinary bodies already embedded in recovered spacetime; ordinary objects occur only in downstream body/readout roles.

Inputs from earlier papers. Paper I supplies the body-first architecture, Paper II the typed finite source support, Paper III the record transport boundary, Paper IV the parent-realization claim boundary, Paper V the clock interface, Paper VI the conditional quantum terminal, and Paper VII-A the separate joint-terminal assembly. These are provenance inputs, not results reproved here. Results owned here. Tau-local standard subsystem composition; the rank-seven support–port reconstruction; the pointed exterior-seed and rank-only no-go theorems; the pair/top exterior no-go, common completion and enriched-class occupation theorem; the operator-square selector; the class-minimal enriched P3 realization and deletion controls; and the frozen public-source eligibility audit.

Claim boundary. “Class-minimal” always means minimal within the declared finite, source-irredundant one-copy completion class. It does not mean that this is the unique or absolutely minimal law of Nature. The paper constructs a sufficient internal realization theorem; it does not derive the existence or occupation of that completion from an unrestricted physical base-seed law. A richer parent may realize the same packet, and an empirical endpoint could demote the proposed law.

Inherited formation basis. The base is the protected zero-source response structure of one joint law, the seed is a response-distinct base-relative source class, and a formed body is a stable selected nonneutral response. This factorization is canonical only after the zero-source member is physically fixed. The rank-seven packet neither fixes that anchor nor derives universal-seed occupation.

Contents

1 Roadmap and ownership discipline	4
2 P3 closure spine	5
2.1 Tau-local origin of standard subsystem composition	5
2.2 One common law for locality and the connected P3 packet	8
2.3 Why the class-minimal law is not a terminal fit	8
2.4 Handoff to the joint-terminal assembly	11
3 Class-minimal enriched-law closure and empirical frontier	11
4 Source origin of the pointed exterior grammar	12
5 Public-source eligibility audit	12
6 Decision ledger	16
7 Rank-seven occupation and finite-local support	16
7.1 Pair/top exterior closure and its exact boundary	16
8 Product-free pair law and updated Nature boundary	17
9 Pointed simple-source compression of terminal fanout	18
10 Observer realization: inherited results and physical boundary	19
11 Source identification and recovery	19
12 Conditional source preparation and time-resolved access	20
13 Conclusion	20
A Detailed P3 typed-role extraction and port geometry	20
B Complete-incidence characterization	32
C Enriched variational ownership and ambient-extension control	33
D Cross-paper operational witness handoff	34

1 Roadmap and ownership discipline

Inherited observer co-descent convention. The rank-seven packet is an upstream source structure, not a detached observer or an independent channel. Operational observerhood additionally requires an occupied body-side carrier, regular local rank-four descent, stable observer–source quantization and a nonzero record/effect. The observer and accessible 4D world are co-readouts; this paper addresses the typed packet that can support that closure and does not claim that rank seven alone creates an observer.

Inherited mixed-Hessian source boundary

This packet enters the post-body response only after its source typing is frozen. Let Y be the occupied source chart, Z the response quotient and

$$H = D_Z^2 \mathcal{A}_{\text{post}} > 0, \quad B = D_Z D_Y \mathcal{A}_{\text{post}}. \quad (1)$$

At a regular stationary response, the implicit-function theorem fixes the complete response germ,

$$\boxed{\Gamma = D_Y Z_* = -H^{-1} B.} \quad (2)$$

For a frozen precursor $F : Y \rightarrow V_F$ and an active response covector p_s , a unique precursor-to-active factor exists exactly when

$$\boxed{p_s \Gamma|_{\ker DF} = 0.} \quad (3)$$

In the inherited SBTCL106 Bregman graph action $\mathcal{A}_{FY} = \frac{1}{2} \|z - J_{OS} y\|_{G_{OS}}^2$, one has

$$H = G_{OS}, \quad B = -G_{OS} J_{OS}, \quad \Gamma = J_{OS}, \quad C - B^* H^{-1} B = 0. \quad (4)$$

Hence no additional mediator action or free solder gain is needed. Zero Schur remainder is graph neutrality, however, not descriptor factorization. Indeed

$$\mathcal{A}_\lambda = \frac{\kappa_d}{2} (c - d)^2 + \frac{\kappa_s}{2} (c - s + \lambda \eta)^2, \quad \kappa_d, \kappa_s > 0, \quad (5)$$

has the same $\eta = 0$ reduct, positive response Hessian and zero Schur remainder for every λ , while $s_* = d + \lambda \eta$ and $c_* = d$. With $F = d$, factorization holds exactly for $\lambda = 0$. Positivity, common mediation and Schur neutrality therefore do not select the vertical source jet. Physical ownership and nonzero occupation of the required J_{OS} component, the complete-terminal fibre test, Q_{OS} and calibration remain outside this paper’s conditional packet theorems.

The proof order is

$$\begin{aligned} \text{support structure} &\longrightarrow \text{rank-seven reconstruction} \longrightarrow \text{variational occupation,} \\ &\longrightarrow \text{Clifford packet} \longrightarrow \text{empirical pair falsifier.} \end{aligned}$$

The first four arrows are finite mathematical statements under explicit premises. The final arrow is an acquisition programme, not a positive Tau signal. Gravity, matter and gauge terminal equations remain owned by Paper VII-A; the scalar-15 and functional-RG specialization is owned by Technical Paper VII-C.

2 P3 closure spine

The detailed P3 typed-role extraction, O/Q and M/G stabilizer analysis and cotangent-port construction are preserved in Appendix A. The first theorem below reconstructs the positive quotient

$$(E_{P3}^{\text{phys}}, K_*), \quad \dim E_{P3}^{\text{phys}} = 7,$$

from a smaller support–port packet. The subsequent representation argument then uses its canonical exterior action

$$\mathbf{C}_{\text{ext}}(v) = \varepsilon(v) + \iota_{K_*(v, \cdot)},$$

and the normalized two-leg record factor. The appendix proves exactly which complete first-jet, metric, typing and operational-port premises are required; it does not import their physical base–seed occupation into the theorems below.

The proof spine should be read in five steps. First, the source-owned support structure isolates two singleton roles and two unresolved role planes. Second, ordered ports and the independent unit reconstruct a typed rank-seven carrier, while the decoder preserves that rank in the body. Third, the class-minimal variational law occupies the reconstructed packet rather than adding occupation as a separate postulate. Fourth, the operator-square and pointed-record terms produce the Clifford–record realization. Finally, the public pair test asks whether its source-whitened anticommutator relation can be evaluated without terminal fitting. Thus the sequence is

$$\begin{aligned} \text{support structure} &\longrightarrow \text{rank-seven reconstruction} \longrightarrow \text{variational occupation} \\ &\longrightarrow \text{Clifford–record packet} \longrightarrow \text{empirical pair falsifier.} \end{aligned}$$

The first four arrows are conditional internal results; the last is the empirical decision frontier and currently remains data-ineligible.

2.1 Tau-local origin of standard subsystem composition

The two-leg record carrier used later should not be inserted merely because standard four-dimensional notation uses a tensor product. Let H_0 and H_1 be finite occupied subhypergraphs of the one universal seed–body realization. Call them *parent relation-free* when no occupied primitive incidence meets both supports. For this scope impose the following finite parent-law clauses:

- (LC1) every primitive closure rule has connected source support contained in one occupied component;
- (LC2) a rule supported in H_i has output support inside the component closure of H_i , so it creates neither a bridge to H_{1-i} nor a shared external mediator;
- (LC3) the primitive component rules exhaust joint admissibility: there is no additional global accept/reject predicate on otherwise lawful endpoint pairs; and
- (LC4) the finite record linearization is source faithful and inherits the source-rigid unrescaled monoidal metric.

The first three clauses concern closure-set factorization. The fourth fixes the common action normalization after factorization; it is not needed for the abstract tensor shape.

Theorem 1 (Tau-local standard subsystem composition). *Under LC1–LC3,*

$$\boxed{\text{Cl}_\tau(H_0 \sqcup H_1) \simeq \text{Cl}_\tau(H_0) \times \text{Cl}_\tau(H_1).} \tag{6}$$

Consequently the finite record algebras obey the canonical isomorphism

$$\boxed{\mathbb{C}^{\text{Cl}_\tau(H_0 \sqcup H_1)} \simeq \mathbb{C}^{\text{Cl}_\tau(H_0)} \otimes \mathbb{C}^{\text{Cl}_\tau(H_1)}.} \tag{7}$$

Local parent operations have the standard commuting form

$$T_0 \mapsto T_0 \otimes I, \quad T_1 \mapsto I \otimes T_1, \quad [T_0 \otimes I, I \otimes T_1] = 0. \quad (8)$$

Under LC4 the isomorphism in Eq. (7) is isometric in the inherited common action unit, with no independent central tensor gain.

Proof. Take any finite global closure sequence starting from $H_0 \sqcup H_1$. LC1 assigns every primitive step to one component, and LC2 keeps its output inside that component. Projecting the sequence onto the two supports therefore gives a pair of lawful local closure sequences and hence a map from the left side of Eq. (6) to the right side.

Conversely, rules on disjoint supports commute: a rule on H_0 changes neither the match nor the output of a rule on H_1 , and conversely. Arbitrary local closure sequences can therefore be interleaved into a lawful global sequence. LC3 forbids a second global condition from deleting the resulting endpoint pair. The two constructions are inverse, proving Eq. (6).

For finite sets A and B , the delta-basis correspondence

$$\delta_{(a,b)} \longleftrightarrow \delta_a \otimes \delta_b \quad (9)$$

extends uniquely to the algebra isomorphism $\mathbb{C}^{A \times B} \simeq \mathbb{C}^A \otimes \mathbb{C}^B$, proving Eq. (7). Component-local rules act on only one delta factor, which gives Eq. (8). Finally LC4 makes the primitive delta bases orthonormal in the inherited source metric and the source-rigid monoidal continuation excludes a separate positive central rescaling. This proves the metric statement. $\square$

Sharp controls and scope. Let both local alternative sets be $\{0, 1\}$ while the global endpoint set is only

$$\mathcal{D} = \{(0, 0), (1, 1)\}. \quad (10)$$

Both marginal projections are onto, yet $\dim \mathbb{C}^{\mathcal{D}} = 2 < 4$; this hidden global relation violates LC3. A crossing rewrite $(i, j) \mapsto (1 - i, 1 - j)$ generates exactly the same diagonal orbit and violates LC1–LC2. Thus disjoint labels, complete marginals, action valuation or a vanishing mixed Hessian do not imply Eq. (6). A positive central metric rescaling preserves the tensor algebra but violates the isometric part of LC4.

The theorem is a Tau-only derivation inside the declared enriched component-local parent class. It does not show that the old-bare admissibility skeleton or the unrestricted physical base-seed law realizes LC1–LC4 on every claimed isolated scope. If a crossing hyperedge is occupied, the relation-free premise fails and this closure-factorization theorem simply does not apply. Retaining a tensor carrier as the kinematic envelope of interacting quantum states then uses the separate strong dagger-monoidal quantum representation result; it does not follow from Eq. (6) alone.

The full seven-role carrier need not be named in advance. Let E_{rel} be the positive six-dimensional relative source carrier and let P_{BMG}, P_{OQ}, P_{RMG} be commuting K_{rel} -self-adjoint projectors. Write E_e for their joint binary eigenspaces.

Theorem 2 (Support–port reconstruction and class-minimal enriched-law realization). *Assume*

$$\dim E_{100} = 1, \quad \dim E_{010} = 2, \quad \dim E_{001} = 1, \quad \dim E_{101} = 2, \quad (11)$$

with every other joint eigenspace zero. Assume further that the two two-dimensional blocks possess source-owned quotient-basic ordered cotangent decompositions

$$E_{010} = L_O \oplus L_Q, \quad E_{101} = L_M \oplus L_G, \quad (12)$$

and that the common-unit vector z_U splits directly from E_{rel} . Then

$$E_{P_3}^{\text{phys}} = L_B \oplus L_O \oplus L_R \oplus L_U \oplus L_Q \oplus L_M \oplus L_G, \quad \dim E_{P_3}^{\text{phys}} = 7, \quad (13)$$

where $L_B = E_{100}$, $L_R = E_{001}$ and $L_U = \mathbb{R}z_U$. If a smooth body realization has a local left decoder, this rank is preserved in the stabilized body.

Normalize the already stabilized body branch by $\hat{\Gamma}_{\text{body}} = \Gamma_{\text{body}} - \Gamma_{\text{body}}[M_*] \geq 0$ on its declared stability basin. Write $\Phi_{\text{port}} = \Gamma_{\text{port}}^{\text{CTC}}$ and normalize the remaining post-body defects by $\Phi_{P_3} = \Gamma_{P_3}^{\text{enr}}/\mathcal{A}_*$ and $\Phi_{\text{rec}} = \Gamma_{\text{rec}}/\mathcal{A}_*$. Define the proposed class-minimal enriched P_3 law within the declared finite, source-irredundant one-copy completion class by

$$\mathcal{L}_{P_3}^{\min} = \left(\mathcal{D}_{\text{SF}}, \hat{\Gamma}_{P_3}^{\min} = \hat{\Gamma}_{\text{body}} + \mathcal{A}_*(\Phi_{\text{port}} + \Phi_{P_3} + \Phi_{\text{rec}}); \text{Real}_B, Q_B, \Omega_{\emptyset}; \text{Phys} = \underset{\mathcal{D}_{\text{SF}}}{\text{Argmin}} \hat{\Gamma}_{P_3}^{\min} \right), \quad (14)$$

where $\mathcal{D}_{\text{SF}}$ owns Eq. (11), the unit split and the two protected cotangent source covectors; $\Gamma_{\text{port}}^{\text{CTC}}$ is the normalized two-block Fenchel–Young law; $\Gamma_{P_3}^{\text{enr}}$ is the once-counted graph plus operator-square law; Γ_{rec} is the normalized two-leg record law; $Q_B \text{Real}_B = I$; and $\Omega_{\emptyset}$ is the pointed cyclic exterior root with the complete physical P_3 null and no bypass. All three Φ terms are nonnegative dimensionless defects in the one common action unit. Then the physical branch selected by Eq. (14) satisfies

$$(B_{\tau}^{\text{phys}}, s_U, \mathcal{L}_{P_3}^{\min}) \implies \mathfrak{P}_{P_3}^{\min}, \quad (15)$$

where $\mathfrak{P}_{P_3}^{\min}$ is the class-minimal faithful typed rank-seven Clifford–record packet, unique up to typed source isometry and typed unitary equivalence in the declared source-irredundant one-copy class.

The earlier narrow reduct does not entail $\mathcal{L}_{P_3}^{\min}$: an equal-upstream no-response completion omits a required ordered port, while dependent-unit and equal-Gram commuting controls independently remove the rank and operator-square conclusions.

Proof. Simultaneous orthogonal spectral decomposition of the three projectors gives the intrinsic blocks $L_B \hat{\oplus} E_{010} \hat{\oplus} L_R \hat{\oplus} E_{101}$. The two ordered cotangent decompositions split the only repeated blocks, and the direct unit split adds one independent line, proving Eq. (13). Differentiating the decoder identity gives $DQ_B D \text{Real}_B = I$, so body realization cannot collapse a typed direction. The two Fenchel–Young graph terms have zero response Schur remainder. The faithful separating operator-square term vanishes exactly on $C(u)C(v) + C(v)C(u) = 2K_*(u, v)I$, and eliminating the graph response leaves that selector once. The pointed exterior root and normalized record factor then give $J_{\text{ext}}^{\dagger} J_{\text{ext}} = I_{256}$ and faithful cyclic occupation. The canonical exterior Clifford incidence, its graph response, the two metric-dual ordered ports, the stabilized body and the normalized two-leg record root form an explicit simultaneous zero of every term in $\hat{\Gamma}_{P_3}^{\min}$. Its global minimum is therefore zero. Since all summands are nonnegative, every global minimizer makes every summand vanish; packet occupation follows from the declared variational law and is not a separate common-zero premise. Complete physical null and no bypass give the stated minimal-class uniqueness. The three cited controls prove non-entailment from the narrow reduct. Appendix A gives the componentwise deletion certificate. QED. $\square$

Equation (15) closes a formal ownership arrow only after the proposed physical law is specified. It does not prove that Nature realizes this proposal, and it does not establish ambient-parent exhaustivity from finite readouts.

2.2 One common law for locality and the connected P3 packet

The locality theorem and the P3 occupation theorem act on different parts of the source decomposition. Let $\pi_0(s_U)$ denote the connected components of occupied primitive incidence and define

$$\boxed{\hat{\Gamma}_{\text{CLP3}}[X] = \sum_{C \in \pi_0(s_U)} \hat{\Gamma}_C[X_C], \quad K_{\text{CLP3}} = \bigoplus_C K_C.} \quad (16)$$

Assume that primitive rules exhaust their full connected supports, preserve component closure, and admit no additional cross-component multiplier or hidden endpoint predicate; that every component action is nonnegative, once-counted and has an attainable zero; and that realization is source faithful in the common unrescaled action unit. On the occupied connected component C_{P3} take $\hat{\Gamma}_C$ to be the already defined $\hat{\Gamma}_{P3}^{\min}$ of Eq. (14).

Theorem 3 (Component-local P3 joint occupation). *Under these hypotheses every global minimizer of Eq. (16) satisfies LC1–LC4 on every parent-relation-free support pair and occupies the complete typed rank-seven packet $\mathfrak{P}_{P3}^{\min}$ on C_{P3} . The earlier narrow base-seed/body reduct does not entail this common law.*

Proof. Full-support assignment and component preservation give LC1–LC2. Primitive exhaustivity and absence of an extra cross-component multiplier give LC3, while source-faithful unrescaled realization is LC4. Hence Theorem 1 applies to every relation-free pair. Nonnegativity, disjoint component variables and attainability give

$$\text{Argmin } \hat{\Gamma}_{\text{CLP3}} = \prod_C \text{Argmin } \hat{\Gamma}_C, \quad \min \hat{\Gamma}_{\text{CLP3}} = 0. \quad (17)$$

Restriction to C_{P3} is therefore a zero of $\hat{\Gamma}_{P3}^{\min}$, so Theorem 2 forces the complete rank-seven Clifford–record packet. Its internal incidences do not violate LC1–LC2: they belong to one connected component and are never factorized by the theorem.

For non-entailment, keep the local rules and complete marginals but impose the hidden diagonal endpoint relation in Eq. (10); LC3 fails. Independently replace the P3 realization by the equal-total-scalar rank-six missing-port control used in the support–port deletion certificate. Their product has the same earlier narrow descriptors while failing both targets. Thus that reduct cannot select Eq. (16). $\square$

This theorem closes compatibility and common-law occupation inside the declared enriched completion. It is not a proof that Nature realizes $\hat{\Gamma}_{\text{CLP3}}$ or that the listed components exhaust the ambient parent.

2.3 Why the class-minimal law is not a terminal fit

Here *class-minimal* means componentwise minimal only within the declared finite, source-irredundant one-copy completion class. In both the law and packet symbols, the superscript “min” has this restricted meaning; it does not assert that the law is Nature’s unique or globally minimal parent law. Nor is the functional fitted backwards from terminal residuals: its inputs are frozen on the source side before endpoint evaluation. The origin of its clauses is as follows.

- (i) The support multiplicities and projectors are inherited from the source-role structure. Their realization by the unrestricted physical base-seed system remains an explicit completion premise.

- (ii) The O/Q and M/G cotangent ports are minimal preimages, within this class, of already required ordered dualities. Deleting either port leaves an explicit typed swap symmetry.
- (iii) The independent common unit and the left decoder have separate rank roles: the unit raises source rank from six to seven, while the decoder prevents collapse in body realization. Neither is a terminal-specific gain.
- (iv) The state-neutral operator-square term is the genuinely new Tau-specific cross-role clause. Without it, a commuting packet can retain the same normalized marginal trace Gram.
- (v) The pointed cyclic exterior root and no-bypass clause are required by the faithful full-rank nontracial and extra-record countermodels. They select record occupation and completeness only within the declared class.
- (vi) The two-leg record tensor is not inferred from a terminal residual. Theorem 1 derives it from the independent component-local no-bridge closure law; the diagonal hidden-relation model prevents replacing that source law by complete marginals or action additivity.

Thus Theorem 2 is a noncircular internal realization theorem for a source-frozen proposed law. It is not a deeper derivation of that law from the narrow parent, and it does not exclude richer parent laws with the same physical zero locus.

Theorem 4 (Pointed exterior-seed closure and rank-only no-go). *Assume that the occupied typed $P3$ seed owns the complete oriented word grammar $\Lambda^\bullet E_{P3}$, explicitly occupies the empty word $\Omega_\emptyset$, and that the base realizes $C_{\text{ext}}(v)$ as in Eq. (78). Assume also that the normalized two-leg Frobenius record root generates an orthonormal record basis r_0, r_1 with no independent bypass. Then the vectors*

$$\Psi_{I,a} = C_{\text{ext}}(e_{i_k}) \cdots C_{\text{ext}}(e_{i_1}) \Omega_\emptyset \otimes r_a, \quad I = \{i_1 < \cdots < i_k\} \subseteq \{1, \dots, 7\}, \quad a \in \{0, 1\}, \quad (18)$$

form an orthonormal basis of $\mathcal{F}_{P3}$. Equivalently, the root-orbit map $J_{\text{ext}} : \mathbb{C}^{256} \rightarrow \mathcal{F}_{P3}$ obeys

$$\boxed{J_{\text{ext}}^\dagger J_{\text{ext}} = I_{256}} \quad (19)$$

Consequently the relation, source-symmetry and cyclic trace–GNS conditions are simultaneous kinematic consequences on this declared carrier; no independent Gram-isometry action term is required there.

Full orbit rank, source nonloss and equal central block weights are not sufficient substitutes. On $\mathcal{A} = M_8(\mathbb{C})^{\oplus 4}$ with normalized trace, choose

$$h_\epsilon = \bigoplus_{b=1}^4 \text{diag}(1 + \epsilon, 1 - \epsilon, 1, \dots, 1), \quad 0 < \epsilon < 1, \quad (20)$$

and $J_\epsilon(a) = a h_\epsilon^{1/2}$ on the regular carrier. Then J_ϵ has rank 256 and equal weight 1/4 on every central block, but

$$J_\epsilon^\dagger J_\epsilon = R_{h_\epsilon} \neq I, \quad \omega_\epsilon([E_{01}^{(1)}, E_{10}^{(1)}]) = \frac{\epsilon}{16} \neq 0. \quad (21)$$

Thus this faithful full-rank root state is nontracial and retains a nonzero metric defect.

Proof. Creation followed by K_* -adjoint contraction sends the empty exterior word to the signed wedge basis vector labelled by the same subset I . Distinct subsets are orthogonal, and every subset occurs once. Tensoring with the two orthonormal record vectors therefore gives exactly $2^7 \cdot 2 = 256$ orthonormal vectors, proving Eq. (19). The Clifford relations give relation generation, the induced exterior action transports every source isometry, and the full word orbit is cyclic; the normalized regular vector state is the represented trace.

For the control, positivity of h_ϵ makes right multiplication by $h_\epsilon^{1/2}$ invertible, so the orbit map has full rank. Every block has trace eight, hence the four central weights are equal. Direct evaluation in the normalized regular inner product gives $J_\epsilon^\dagger J_\epsilon = R_{h_\epsilon}$. Finally $[E_{01}, E_{10}] = E_{00} - E_{11}$, whose h_ϵ -weighted normalized trace in one of the four blocks is $2\epsilon/32 = \epsilon/16$. The root state is therefore not tracial and its Gram cannot be the source identity. QED. $\square$

The positive half of Theorem 4 is conditional on physical ownership of the complete pointed exterior grammar. The negative half is an exact countermodel: dimension, rank and the retained discrete block symmetries cannot replace that source-owned pointed metric. The unrestricted physical base-seed law has not yet been shown to own this grammar.

Appendix B packages the same requirement as an exact complete-incidence characterization: within the declared finite source-irredundant Clifford-recovery class, a minimal faithful typed P3 subpacket exists uniquely up to typed unitary equivalence if and only if the parent owns the complete rank-seven incidence. This is a characterization of the missing object, not a proof of its unrestricted physical existence.

The certificate also admits one finite positive source selector. On the dimensionless normalized quotient $(E_{P_3}^{\text{phys}}, K_*)$, let $n = \dim E_{P_3}^{\text{phys}} = 7$, let $C : E_{P_3}^{\text{phys}} \rightarrow \mathcal{A}_{\text{body}}^{\text{sa}}$ be linear, and let τ be a faithful normalized trace. Define

$$\mathcal{V}_{\text{OS}}[C] = \frac{\mathcal{A}_*}{2} \int_{S(E_{P_3}, K_*)} \tau \left[(C(v)^2 - I)^\dagger (C(v)^2 - I) \right] d\mu(v), \quad (22)$$

where $d\mu$ is normalized orthogonal-invariant sphere measure. Before unit-sphere restriction the defect is $C(v)^2 - K_*(v, v)I$, so the common action unit $\mathcal{A}_*$ restores dimensions without a role-dependent gain. For a K_* -orthonormal frame, write $A_a = C(e_a)$, $d_a = A_a^2 - I$, and $h_{ab} = A_a A_b + A_b A_a$. The fourth sphere moments give the exact finite formula

$$\frac{2\mathcal{V}_{\text{OS}}[C]}{\mathcal{A}_*} = \frac{1}{n(n+2)} \left[2 \sum_a \|d_a\|_\tau^2 + \left\| \sum_a d_a \right\|_\tau^2 + \sum_{a < b} \|h_{ab}\|_\tau^2 \right]. \quad (23)$$

Theorem 5 (Operator-square source selector). *For a faithful normalized trace, $\mathcal{V}_{\text{OS}}[C] \geq 0$, with equality if and only if*

$$C(v)^2 = K_*(v, v)I \quad \text{for every } v \in E_{P_3}^{\text{phys}}. \quad (24)$$

Equivalently, $C(u)C(v) + C(v)C(u) = 2K_(u, v)I$. The zero set is invariant under every K_* -orthogonal source-frame change.*

Proof. The integrand in Eq. (22) is continuous and nonnegative. Faithfulness of τ makes a zero integral equivalent to a zero operator defect on the unit sphere; homogeneity extends the result to all v . Polarization gives the anticommutator, and the reverse implication is immediate. Orthogonal invariance follows from invariance of K_* and $d\mu$. Finally, $\mathbb{E}[v_a^4] = 3/[n(n+2)]$, $\mathbb{E}[v_a^2 v_b^2] = 1/[n(n+2)]$, and the odd moments vanish, yielding Eq. (23). $\square$

This selector contains information absent from the scalar Gram matrix. The canonical Clifford family and seven commuting four-qubit Pauli-Z words act on the same 16-dimensional carrier and both obey $\tau(A_a A_b) = \delta_{ab}$. The finite certificate nevertheless gives

$$\frac{2\mathcal{V}_{\text{OS}}^{\text{Clifford}}}{\mathcal{A}_*} = 0, \quad \frac{2\mathcal{V}_{\text{OS}}^{\text{commuting}}}{\mathcal{A}_*} = \frac{4}{3}. \quad (25)$$

Thus scalar Hessian/Gram data cannot select the physical packet, whereas the operator-valued source term can.

Appendix C gives the staged enriched-parent functional whose nonnegative zero locus combines this operator-square law with the P3 handoff without double counting. That theorem proves ownership inside the declared enriched completion. It does not prove that the unrestricted physical base-seed law realizes or occupies the enrichment.

2.4 Handoff to the joint-terminal assembly

The results above close only the algebra/representation level. Foundation Paper VII-A separately owns the once-counted terminal functional, the Born effect derivative and the conditional Einstein–Maxwell–matter equations. Combining VII-A with Theorem 2 is therefore a cross-paper dependency composition, not another theorem proved in this technical paper. In particular, rank-seven typing or nonzero activation must not be mistaken for unrestricted physical occupation of the complete packet.

3 Class-minimal enriched-law closure and empirical frontier

The earlier narrow reduct has the exact non-entailment

$$(B_{\tau}^{\text{phys}}, s_U, \mathcal{L}_{\text{narrow}}) \not\Rightarrow \mathfrak{P}_{P3}^{\text{min}}. \quad (26)$$

The no-response port control already proves Eq. (26); the dependent-unit, equal-Gram commuting and nonracial-root controls locate three further independent failures. It is therefore impossible to remove this no-go by another manipulation of the unchanged reduct.

Theorem 2 instead supplies an explicit proposed physical law and proves

$$(B_{\tau}^{\text{phys}}, s_U, \mathcal{L}_{P3}^{\text{min}}) \Rightarrow \mathfrak{P}_{P3}^{\text{min}}. \quad (27)$$

Thus the formal parent-law realization and occupation arrow is closed inside the declared enriched physical MVP. The phrase “Nature realizes this law” is not an additional conclusion that can be proved from an unspecified parent: it is the empirical status of the proposed law. Likewise, Eq. (27) does not establish that the selected connected packet exhausts every ambient source extension.

Two scientific routes remain, but they have different status:

1. *Deeper-origin route.* Derive $\mathcal{L}_{P3}^{\text{min}}$ from a more primitive enriched base-seed law. This could strengthen the theory, but is no longer required for the internal consistency of the stated completion; Eq. (26) forbids obtaining it from the unchanged narrow reduct. The standard subsystem-composition part of this route is now closed conditionally by Theorem 1; the remaining deeper-origin debt concerns physical realization of LC1–LC4 and of the full support–port, operator-square and pointed-root law.
2. *Discriminator route.* Freeze a source-side morphological descriptor and derive a gain-free, parameter-free relation between at least two terminal readouts that is not implied by the corresponding standard comparator. The immediate algebraic target is $\{C_a, C_b\}/2 = \delta_{ab}I$ in a source-whitened frame; the stronger $\mathcal{O}_{P1|P3} = 0$ requires an independently certified global source-completeness premise. Any endpoint prediction must be fixed before data are scored.

Failure of a source-frozen discriminator demotes the proposed physical law. Neither route may be repaired by adding another standard-limit recovery or by selecting source structure from terminal residuals.

4 Source origin of the pointed exterior grammar

Theorem 6 (Conditional exterior-seed selection). *Let V_s be the finite primitive possibility space owned by the seed. Assume that its composition is unital, bilinear, associative and oriented; repeated occupation of one primitive direction is excluded, $v^2 = 0$; and no relation other than those forced by these premises is imposed. Then the complete unital source grammar is*

$$\mathcal{G}_s \cong \Lambda^\bullet V_s.$$

The empty word is the distinguished root. If the base supplies a positive metric on V_s , insertion and contraction generate the corresponding finite Clifford action and the pointed cyclic carrier required by the rank-seven packet.

Proof. Polarizing $(u + v)^2 = 0$ gives $uv = -vu$. The universal unital associative algebra with precisely these relations is the exterior algebra. Its degree zero element is the unique empty word. A positive metric identifies V_s with its dual, so exterior multiplication and contraction obey the Clifford anticommutator. Acting on the empty word generates the full exterior carrier. $\square$

The tensor algebra is the control obtained by dropping multiplicity exclusion; the symmetric algebra is the control obtained by replacing orientation with commutation. Hence primitive inventory and composition alone do not force this theorem. It identifies the weakest declared source class that realizes the pointed packet, while physical occupation by the unrestricted base-seed parent remains open.

5 Public-source eligibility audit

The accompanying reproducibility package freezes a finite source search as of 7 August 2026. A candidate is eligible for a future independent action-quantum endpoint score only if all six predeclared gates hold:

1. an independently reconstructed electromagnetic or morphological leg;
2. quantum-state coherence, phase or tomography;
3. same-system joint indexing;
4. an independent positive endpoint action, energy or work measurement;
5. a complete standard comparator; and
6. a candidate-specific operational realization of the source-frozen, gain-free Tau cross-readout law on independently measured observables.

The script `scripts/audit_public_source_eligibility.py` contains all 35 candidate identifiers, public links, gate values and textual verdicts and regenerates `data/derived/public_source_eligibility_audit.json`. Zero of 35 candidates pass all six gates. The gate-pass counts in the listed order are (30, 25, 22, 11, 35, 0). After Theorem 2, the abstract Tau law itself is frozen; the last zero now records the absence of a candidate-specific independent operational map from the common physical source into its typed observables and whitening metric. Even before that operational gate is applied, zero candidates pass all five data/comparator gates; the nearest candidates pass four. Thus the endpoint-data packet and its common-source realization remain independently incomplete. This finite audit is a reproducible classification of the located sources, not an exhaustive theorem about all existing experiments; individual gate assignments remain open to correction when stronger public metadata or raw records appear. Failure of source eligibility is an acquisition and identifiability limitation, not a null Tau signal. The frozen query families,

inclusion rule, source-version policy and the absence of external payload hashes are recorded in `data/public_source_search_protocol.md`.

Consequently the direct rank-seven Nature test is frozen but not yet executable: no primary residual is scored on an ineligible packet. The available Wen path-phase reconstruction is only an auxiliary branch control; its morphology increment beyond scalar action is not significant under the shared-kernel null ($\Delta R^2 = 0.1758$, $p = 0.144$) and does not test or falsify the complete P3 relation.

Minimal two-role falsifier. A first rejection does not require simultaneous reconstruction of all seven roles. For any predeclared distinct typed pair (a, b) , the proposed minimal P3 law necessarily restricts to

$$C_a^2 = C_b^2 = I, \quad \{C_a, C_b\} = 0 \quad (28)$$

after the same source-frozen whitening used by the full test. Therefore a statistically resolved nonzero anticommutator on one eligible pair rejects the minimal rank-seven completion. Eligibility still requires two independent typed source directions and an informationally complete reconstruction of the symmetrized operator product; a single-state expectation value is not enough. A passing pair does not establish the remaining roles or rank seven. The reproducible focused audit in `scripts/audit_minimal_pair_nature_test.py` finds zero eligible packets among 31 targeted public architectures. This 0/31 result narrows the acquisition requirement and supplies no Tau pass or failure.

There is one exact product-free consequence. For every common prepared state, Eq. (28) implies

$$\langle C_a \rangle^2 + \langle C_b \rangle^2 \leq 1. \quad (29)$$

For every real unit vector n , the operator $n_a C_a + n_b C_b$ is an involution, so positivity bounds its expectation by one; maximizing over n proves Eq. (29). A commuting control has a joint $+1, +1$ state and gives two. Thus one resolved value above one falsifies the minimal completion without product tomography. A value below one does not confirm Tau, because an apparatus-selected anticommuting Pauli pair satisfies the same standard-QM identity. The maps and their orientation must be reconstructed independently of the target relation.

The already derived ROOT-TT same-preparation identity is not an independent realization of this pair. At frozen occupation and overlap it has the form $T_{TT} = N_O K \rho_{01}$, so its real two-component Jacobian is an invertible rescaling of the ROOT-coherence Jacobian. The stacked Jacobian therefore has rank two, equal to either complex terminal separately, and adds zero cross-terminal novelty rank. This factorization is a physical cross-terminal prediction, but counting it as two primitive source directions would duplicate one source jet. The executable test consequently requires two independently controlled typed perturbations and process or sequential data that reconstruct their symmetrized cross-response.

Nor can complete tomography confined to the quantum terminal replace the missing typed leg. If $L_1 = A_1 L_Q$ and $L_2 = A_2 L_Q$, the two maps may describe independent quantum gates or coordinates, but both still descend through the same Q role. Relabelling one as M or G after endpoint inspection would violate source freeze. The audited higher-order Quantum SWITCH process matrix and donor-spin gate-set tomography packages are therefore near candidates, not eligible cross-role packets: both reconstruct rich quantum operations but publish no independently sourced metric, stress or morphology perturbation.

A closer hybrid candidate is a public source-data package in which a superconducting qubit is read through a piezo-opto-mechanical transducer. It contains transduction characterization, prepared-qubit IQ records, optical-power scans, and pump-dependent T_1/T_2^* families on one device. The release nevertheless separates these into figure-level aggregate families: it provides neither an

independently reconstructed mechanical/stress state nor shared record keys for an informationally complete quantum–mechanical symmetrized product. Optical-pump dependence is therefore a controlled environmental response, not yet a second primitive typed map. The package is an acquisition template, not an executable P3 pair test.

A mechanical carrier does not by itself evade the same-sector no-go. If $L_{\text{state}} = A_s L_Q$ and $L_{\text{mech}} = A_m L_Q$, both reconstructions descend through one quantum role even when the carrier is a mechanical oscillator. A calibrated displacement control is likewise not an independent output map. The audited ETH motional-grid-qubit release contains rich characteristic, Wigner, Pauli and process tomography, but all records reconstruct one motional quantum carrier Fluehmann et al. [2019]. The NIST amplified-motion release is closer because it joins calibrated displacement, squeezing and spin response on common parameter rows; nevertheless the displacement is a control and the oscillator output is inferred through the spin/Q readout Burd et al. [2019]. Neither release supplies an independently reconstructed mechanical or stress output whose differential adds rank after conditioning on L_Q , nor the required symmetrized cross-product record. One physically close public candidate couples a membrane oscillator to a distant atomic-spin oscillator and reconstructs a joint Gaussian EPR witness Thomas et al. [2021]. Its released source files expose one combined optical trajectory or spectrum. The separately named mechanical, atomic and backaction spectra are fitted model components rather than independently measured output maps. The package therefore does not certify typed rank two or the informationally complete symmetrized product.

The extended census also includes the Novikov atom–light EPR network Novikov et al. [2025]. Two homodyne detectors record the optical signal and idler arms, but the proposed force sensor is not present in the realized signal arm and the spin response is encoded in the idler photocurrent rather than independently measured by a spin/body terminal. Its open data and code are a strong hybrid-channel precedent, not a typed rank-two P3 packet.

Federated mosaic identifiability no-go. Several incomplete experiments can constrain one common source Gram only after independently certified source-frozen transports place their measured objects in the same carrier representation and action unit. Let $G \in \mathcal{G}$ be the declared source-whitened Gram, let $g = \text{vech } G$, and write the constraints from packet k as $y_k = A_k g$. For $A = \text{stack}_k A_k$, the Gram is unique exactly when

$$\ker A \cap \text{vech}(\mathcal{G} - \mathcal{G}) = \{0\}. \quad (30)$$

The reproducible audit in `scripts/audit_federated_gram_completion.py` applies this criterion to the fourteen frozen pair candidates. Even optimistically granting both normalized Q and M marginals, the design for $g = (G_{QQ}, G_{MM}, G_{QM})$ has rank 2/3 and leaves G_{QM} free. All positive-semidefinite matrices

$$G(\rho) = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad -1 \leq \rho \leq 1, \quad (31)$$

therefore reproduce those marginals. In the seven-role control, the seven diagonals plus a connected six-edge overlap tree have rank 13/28 and leave fifteen symmetric-Gram directions unobserved. Connectivity alone is not identifiability; fewer than complete pair coverage can suffice only under additional algebraic or low-rank identities derived independently of the endpoint data.

Scalar Gram closure is itself only necessary. The anticommuting pair (Z, X) and the commuting pair $(Z \otimes I, I \otimes Z)$ have the same identity normalized-trace Gram, while their anticommutators differ. Thus no synthetic Nature packet or Tau score may be formed by stacking unrelated marginals. A valid mosaic still needs common-carrier transport certificates, at least one independently measured cross-role edge for the scalar pair, and process or sequential tomography for the operator

product. The finite verdict is `FEDERATED_MARGINALS_DO_NOT_IDENTIFY_CROSS_ROLE_GRAM`; it is an identifiability no-go, not a null Tau result.

The reproducibility package is not limited to claim-string checks. The script `scripts/audit_theorem_certificates.py` independently computes finite witnesses for the Schur response, the nonzero invariant-potential minimum, the ROOT-TT Jacobian rank, overlapping terminal maps, the distinction between activation and full occupation, and the normalized same-reduct counterpair. These calculations certify representative finite instances; the analytic proofs remain the source of the universal statements. The separate script `scripts/audit_rank7_parent_law.py` reconstructs the support multiplicities, the successive stabilizer reductions, the independent-unit and decoder rank controls, and the Clifford/equal-Gram counterpair. It also verifies a simultaneous zero of the normalized nonnegative action components, which is the finite certificate for the variational-occupation step in Theorem 2. The federated audit independently verifies the pair and seven-role design ranks, positive-semidefinite completion interval, resolution stability and the equal-Gram commuting operator countermodel; all thirteen checks are generated from the packaged pair ledger.

Code and data availability. The manuscript source, deterministic audit scripts, tests, derived JSON ledgers and the frozen public-source search protocol are distributed in the reproducibility archive accompanying this manuscript. No observational endpoint data are redistributed or fitted here; external candidate records are identified by public links in the frozen audit ledger. Running `python3 scripts/reproduce.py` rebuilds the figures, theorem witnesses, PDF, arXiv source archive and test suite from the packaged files.

6 Decision ledger

Statement	Status	Boundary
Component-local, no-hidden-relation closure linearizes to the standard tensor product	proved conditionally	exact controls show why local marginals or additive action alone are insufficient
Support, two ordered ports, a common unit and a decoder reconstruct rank seven	proved in the declared class	deletion controls establish componentwise minimality only in that class
Pointed exterior seed fixes the cyclic Gram packet	proved conditionally	a faithful nonracial control defeats rank-only selection
Existing top-form closure selects the axial pair closure	false	$\text{Hom}_{S_4}(V_{\text{ax}}, \Lambda^3 V_{\text{ax}}) = 0$
One full-rank seed incidence supplies pair and top closure	proved in the declared exterior class	one metric and one action unit, with ranks $3 + 1$
Pointed-exterior source-fidelity/Hodge law occupies both degrees	proved in the declared enriched class	the narrow base-seed reduct does not entail that grammar or its occupation
Operator-square functional selects the Clifford law	proved	scalar Gram agreement alone does not select it
The enriched law occupies the packet	proved by a simultaneous global zero	class-relative law-to-packet realization
The narrow reduct entails the enriched law	disproved	exact same-reduct counter-models
The unrestricted physical base-seed parent realizes this law	open	must be derived from deeper physics or tested by a source-frozen discriminator
Current public data execute the discriminator	false in the frozen audit	acquisition limitation, not a Tau falsification

7 Rank-seven occupation and finite-local support

The rank-seven packet is physical only through a nonzero effective source, not through algebraic availability alone. For any finite smooth local extra irrep,

$$e_{*,\lambda} = (C_\lambda - B_\lambda^\dagger A^{-1} B_\lambda)^{-1} (k_\lambda - B_\lambda^\dagger A^{-1} j).$$

Positive unloaded split irreps vanish, while stable loaded or cross-coupled irreps are generated support. Inside the declared enriched class, the simultaneous zero of the class-minimal law provides the required load and occupation. This does not prove that the unrestricted physical base-seed parent realizes that law or exclude global or rank-novel ambient extensions.

The upstream operational identity sharpens the last clause. If the complete pointed-exterior grammar and its joint state are constant on complete seed fibres, the enriched one-source premise places the occupied rank-seven packet inside the seed-generated physical quotient. A completely inert ambient summand is then presentation surplus. A loaded or readout-effective rank-novel sector at fixed complete seed key is instead a second source and falsifies the one-source completion. The present narrow reduct does not prove the required seed-fibre constancy.

7.1 Pair/top exterior closure and its exact boundary

Let E_{src} and E_{ax} be three-dimensional oriented carriers and let $J_s : E_{\text{src}} \rightarrow E_{\text{ax}}$ have rank three. The available ternary closure is a top-form line, whereas the oriented pair defect transforms as an

axial three-dimensional representation. Therefore

$$\text{Hom}_{S_4}(V_{\text{ax}}, \Lambda^3 V_{\text{ax}}) = 0,$$

so an equivariant linear use of the top-form closure cannot produce the pair closure. This is a representation no-go, not a missing normalization.

The minimal common typed completion is

$$W_{\text{cl}}^{\text{pre}} = \Lambda^2 E_{\text{ax}} \oplus \Lambda^3 E_{\text{ax}}, \quad K_{\text{cl}} = \Lambda^2 J_s \oplus \Lambda^3 J_s.$$

Invertibility of J_s gives ranks three and one in the two blocks. With the exterior metrics induced by one base metric, the once-counted action

$$\mathcal{A}_{\text{cl}}^{\text{ext}} = \frac{\mathcal{A}_*}{2} (\|(\Lambda^2 J_s)^* \omega_2\|^2 + \|(\Lambda^3 J_s)^* \omega_3\|^2)$$

has Hessian

$$H_{\text{cl}}^{\text{ext}} = \mathcal{A}_* K_{\text{cl}} K_{\text{cl}}^* \succ 0.$$

No degree-dependent gain appears. The completion is componentwise minimal in this graded class: deleting Λ^2 removes all three pair directions, whereas deleting Λ^3 removes the top-form direction.

Occupation is stronger than availability. If a pointed seed owns $\Lambda^\bullet E_{\text{src}}$ with occupied root, and a faithful Hodge base with a source-fidelity/no-bypass law selects J_{ext} , then

$$J_{\text{ext}}^* J_{\text{ext}} = I, \quad \text{rank}(P_2 J_{\text{ext}}) = 3, \quad \text{rank}(P_3 J_{\text{ext}}) = 1.$$

Equivalently, for $\mathcal{D}_{\text{ext}}(J) = \|J^* J - I\|_{\text{HS}}^2$, deletion of the degree-two image costs three missing unit directions and deletion of degree three costs one, while the faithful realization has zero cost. Thus both closures are jointly occupied in the declared connected enriched class.

The narrow relation/order seed reduct does not own the complete pointed exterior state-and-possibility grammar, and therefore does not entail this occupation. A same-local-reduct component control can retain or delete one degree. The result closes the enriched-class realization while leaving unrestricted physical base-seed ownership open.

Global-law proof obligation

The class-minimal rank-seven packet is not by itself a theorem of global physical occupation. The source-fidelity/Hodge law forces every positive-Schur additional noncontractible sector to have $k_{\text{eff}} = 0$, leaving the seed packet uniquely occupied. However, the common local reduct cannot prove that Nature owns this law because an independent component zero-jet can reverse the global minimum. Physical promotion requires an independent source derivation or a source-frozen discriminator, not another internal rank or Clifford closure.

8 Product-free pair law and updated Nature boundary

The minimal pair law has a necessary consequence that does not require tomography of the symmetrized product. For any normalized positive state and any independently reconstructed occupied pair,

$$\boxed{\langle C_a \rangle^2 + \langle C_b \rangle^2 \leq 1.} \tag{32}$$

Indeed, anticommutation and unit squares imply $(n_a C_a + n_b C_b)^2 = I$ for every real unit vector n . The expectation of that involution has magnitude at most one, and maximization over n proves

Eq. (32). A commuting pair has a joint $+1, +1$ state and attains two, so a resolved value above one falsifies the minimal completion. A value at or below one does not confirm it: a Pauli pair selected by the apparatus satisfies the same law by standard quantum mechanics. The typed maps and their orientation must be reconstructed independently of the target Clifford relation.

The extended executable census is updated through 27 August 2026. It audits 31 public or publication-level architectures and finds zero packets with two independently calibrated source-whitened typed maps of stacked rank two on one preparation family. The new Novikov atom–light EPR network Novikov et al. [2025] has two homodyne records and open data/code, but the proposed force sensor is not realized in the signal arm and the spin response is encoded in the idler optical terminal rather than measured by an independent body terminal. The result is

PD-USS1 conditionally selects, Nature occupation not yet identified.

PD-USS1 is owned upstream by Foundation Paper IV. It does not remove this measurement obligation, and PD-USS-N1 proves that selection alone does not force a standard-leaf-orthogonal signal.

9 Pointed simple-source compression of terminal fanout

The pointed exterior carrier also compresses the downstream quantum–gravity fanout premises. On its onto no-bypass CAR orbit, creation and contraction monomials generate the full matrix algebra. Hence the empty root is cyclic, the represented source module is absolutely irreducible and the physical support contains no spectator copy. If the complete source representation additionally owns a pointed oriented two-role plane with nonzero Q and G idempotents, each role fibre contains exactly one equivalent source copy. The inherited common product metric then supplies the equal pullbacks needed for the normalized Schur fanout theorem.

This is a strict reduction, not parent selection. Without the typed role plane, any nontrivial $U(2)$ rotation of that plane commutes with the untyped source action and preserves its scalar metric while changing the named Q and G projectors. A zero role leg remains equivariant, and an extra role copy restores multiplicity freedom. Thus the class-minimal pointed exterior law removes cyclicity, irreducibility and one-copy support as independent debts, but it does not derive the nonzero oriented Q/G incidence from the pointed source alone. Within the declared enriched P3 law the two existing nonzero ordered cotangent ports already have source-owned Q and G types. Their metric duals form the oriented role plane, and LC1–LC4 factorization with the pointed source produces exactly one source copy in each role, so no separate connector or relative gain is required. This conclusion is sharp: the hidden-diagonal subspace has full Q and G marginal ranks but only one globally constrained source copy and is not invariant under the separate role projectors. The remaining unrestricted question is whether the physical base–seed parent jointly realizes the enriched port packet and this Tau-local factorization.

That joint realization already has an explicit once-counted completion inside the declared class. The component-local enriched action $\widehat{\Gamma}_{\text{CLP3}} = \sum_C \widehat{\Gamma}_C$ assigns every primitive defect to one full connected support, contains no hidden global endpoint predicate or cross-component gain, and uses the class-minimal P3 law on its connected P3 component. Attainable nonnegative component zeros make every global minimum occupy both LC1–LC4 and the complete P3 packet. The port and pointed-CAR theorems therefore close the unique isometric Q/G fanout, with $\mathcal{A}_* = \hbar$ on the faithful phase branch. Hidden-global-relation and missing-port controls with the same narrow reduct prove that this is a completion theorem, not a derivation of Nature’s law from the narrow parent. The theorem also identifies only a physical completion class, not a unique off-shell functional. The family $\Gamma_{\text{CLP3}} + \lambda \Gamma_{\text{CLP3}}^2$ has the same minimum set, true-normal physical Hessian and typed packet for

every $\lambda > 0$, and hence the same fanout. Exact representative selection is therefore unidentifiable and unnecessary; physical selection means membership in this class.

10 Observer realization: inherited results and physical boundary

The current observer programme distinguishes a conditional mathematical construction from an occupied physical measuring system. The source order remains base plus one universal seed, stabilized morphological response, observer–source restriction, shared descriptor, and resolved terminal readouts. Neither a detector nor an empirical residual constructs the parent body backwards. Dynamics below concern downstream subsystems only.

The canonical observer-origin derivation, Sections 22–26, owns the following results. The existing joint source action fixes static and frequency-dependent response together. Variation of the BRAC potential couples to the field square; a linear field coupling is a distinct comparator unless derived by expansion about an occupied background. Fixed canonical means select a unique coherent state in a finite positive harmonic packet only if there is no excess fluctuation energy. This is a free/asymptotic preparation statement, not the ground state of an interacting detector. For the existing coupled harmonic pair, the joint ground is pure but either local subsystem is mixed, with its covariance fixed by the complete joint Hamiltonian.

These are conditional derivations and internal numerical checks. The actual observer–source context, physical preparation process, kinetic/energy calibration and stable operational resolution remain unproved. The existing ORA-T1 selector conditionally chooses stable quantizers when the occupied context, admissible partition family and positive closure barrier are supplied; it does not establish those physical premises. A computed ground state is not a preparation mechanism, and a covariance or optimal abstract measurement is not an implemented stable quantizer Q_{OS} .

Ownership and use. The central observer-origin note owns the new contact, state and covariance derivations; the body-readout technical companion carries the consolidated mathematics. Foundation Papers IV, V and VI retain physical-selection, clock-descent and quantum-readout ownership, respectively. This paper imports only the consequences within its scope. No observational dataset, baseline, frozen coefficient, score or empirical conclusion is revised by this update. The reproducibility supplement records the source hashes and the exact conditional/physical distinction.

Consequence for this paper. These results do not select a particle spectrum, coupling constant, Higgs vacuum or gravitational terminal. Common source ancestry remains weaker than physical realization of those sectors.

11 Source identification and recovery

The current observer audit separates a conditional forward construction from physical source selection. A frozen body, occupied observer incidence, preparation, complete instrument and stable resolution can determine records without selecting those inputs from Nature. Equal effects do not fix postmeasurement states, and symmetry within a chosen interaction family does not exclude other symmetric couplings. These results do not select a particle spectrum, coupling constant, Higgs vacuum or gravitational terminal. Common source ancestry remains weaker than physical realization of those sectors.

Dependency and claim ownership. Foundation VI and the body-readout technical companion own the finite instrument/access proof; Foundation IV and TD-0 retain physical source selection; Foundation V, the temporal synthesis and seed-body companion own the gradient/inertial distinction. This paper inherits only the stated specialization. Canonical assumptions, countermodels and calibration audits are in https://github.com/tau-core-research/tau-core-theory/blob/main/docs/tau_rqm_instrument_to_record_bridge_001.md. No physical seed, Nature occupation or Tau-exclusive observation is established by this update. Existing empirical endpoint definitions and scores are unchanged.

12 Conditional source preparation and time-resolved access

The common-operator scalar-Dirac completion now supports an explicit conditional chain from interaction vertices to pair transfer and joint readout. Nonzero vertices do not select occupation. Relational coherence can be generated from an occupied number-diagonal pair even while both local mean fields vanish. In a declared resonant rotating-wave and Markov loss model, finite-window emission retains this coherence information, whereas an all-time count discards it. This is standard-compatible quantum recovery, not a Tau-specific observational residual. These results do not select a particle spectrum, coupling constant, Higgs vacuum or gravitational terminal. Common source ancestry remains weaker than physical realization of those sectors.

Dependency ownership and limits. The P4 common-source assembly owns the scalar/Yukawa benchmark and finite preparation calculations; Foundation VI and the body-readout companion inherit the joint-access mathematics. Foundation IV and TD-0 retain physical source selection, and Foundation V retains clock descent. The parent remains atemporal: the evolution above is a conditional downstream 4D model, not parent memory or a derived morphogenetic time. Physical initial occupation, a gauge-complete realization, detector amplification, absolute calibration and stable observer-relative Q_{OS} remain unselected. No empirical endpoint or score is changed. The canonical derivation and reproducible numerical sources are linked at https://github.com/tau-core-research/tau-core-theory/blob/main/docs/tau_core_p4_nonabelian_particle_common_source_assembly_001.md.

13 Conclusion

The missing object is no longer an unspecified “complete rank-seven incidence.” The declared completion has an explicit support-port preimage, a canonical cyclic record carrier, an operator-square selector and a nonnegative class-minimal law whose minimum realizes and occupies the packet. The accompanying controls show separately why rank, central weights, local marginals, action additivity and the unchanged narrow reduct do not suffice.

This is a conditional internal closure, not unrestricted parent selection. The decisive next result must therefore be either a deeper physical derivation of the enriched completion class from the actual base-seed response or a source-frozen, gain-free cross-terminal prediction that is not implied by standard physics. No additional standard-limit recovery can substitute for either result.

A Detailed P3 typed-role extraction and port geometry

This appendix preserves the complete P3 extraction argument while keeping the main proof spine visible. It contains the detailed propositions for the squared-Hodge realization, the complete first-

jet quotient, source-rank and typed-extraction tests, represented-role and common-scale candidates, the O/Q and M/G stabilizer analysis, and the cotangent-port construction. None of these technical reductions proves that the unrestricted physical base-seed parent occupies the required complete typed incidence.

Proposition 1 (Full-role squared-Hodge selector and P3 realization). *Let E_{P3} be the seven-role physical quotient with positive metric K_* . If one source-owned linear self-adjoint operator map $C : E_{P3} \rightarrow \mathfrak{A}_{\text{src}}$ satisfies*

$$C(v)^2 = K_*(v, v)I \quad \text{for every } v \in E_{P3}, \quad (33)$$

then

$$C(u)C(v) + C(v)C(u) = 2K_*(u, v)I. \quad (34)$$

Thus an orthonormal seven-role frame generates $\text{Cl}_7(\mathbb{C}) \simeq M_8(\mathbb{C}) \oplus M_8(\mathbb{C})$. In the canonical eight-qubit Jordan–Wigner embedding into the common $\text{Cl}_{16}(\mathbb{C}) \simeq M_{256}(\mathbb{C})$ envelope, the two simple sectors occur with multiplicities $(16, 16)$ and normalized-trace weights $(1/2, 1/2)$.

Proof. Apply Eq. (33) to $u + v$ and subtract its u and v instances to obtain Eq. (34). The complex Clifford classification gives the two simple rank-eight sectors. The canonical first seven Jordan–Wigner generators have 128 linearly independent Clifford words and central volume eigenvalues of opposite sign with equal eight-dimensional multiplicity in the minimal faithful direct sum. Sixteen copies embed this faithful representation into dimension 256, giving the stated multiplicities and trace weights. These finite checks are regenerated by `scripts/audit_theorem_certificates.py`. $\square$

The operator-square premise is stronger than a normalized trace-Gram or scalar squared-action identity. Seven commuting diagonal matrices can have the same identity trace Gram as seven Clifford generators while their generic linear combination does not square to a scalar. Hence the P3 packet is an explicit coefficient-free completion and Eq. (33) is an exact selector, but the unrestricted base-seed reduct has not selected the required typed seven-role operator incidence as part of one occupied packet. The existing rank-two EACER and rank-nine seed/context squared-Hodge carriers have the wrong domains. A post-body Clifford handoff is therefore an admissible minimal working completion, not an unrestricted base-seed derivation or a modification of the already stabilized body.

There is a further type distinction. The seven entries in the P3 support ledger are unital role algebras, not seven canonically one-dimensional source slots. Their units are identified in the unital primitive colimit and hence span only one common line. Selecting one non-scalar element inside each role algebra would add noncanonical data and need not be invariant under internal role automorphisms. Therefore the support ledger and the current colimit do not by themselves construct the seven-dimensional domain used in Proposition 1.

The sufficient source object can be stated without choosing such generators. Let $(\mathcal{P}_{P3}^1, G)$ be the complete de-duplicated real physical first-jet module after the true-null quotient, and suppose the source owns a surjection

$$q_{P3} : \mathcal{P}_{P3}^1 \longrightarrow E_{P3}^{\text{lbl}} \simeq \mathbb{R}^7 \quad (35)$$

whose seven typed rows are nonzero, whose kernel is annihilated by every claimed P3 first-jet terminal, and through which every relevant degree-one P3 response factors. If the common action unit normalizes it as

$$q_{P3} G^{-1} q_{P3}^* = I_7, \quad (36)$$

then its metric adjoint $j_{P_3} = G^{-1}q_{P_3}^*$ is a canonical isometric section:

$$q_{P_3}j_{P_3} = I_7, \quad j_{P_3}^*Gj_{P_3} = I_7. \quad (37)$$

The exterior construction on $E_{P_3}^{\text{lbl}}$ then supplies the normalized Clifford law of Proposition 1. This is a conditional role-quotient theorem, not a derivation of Eq. (35) from the physical base-seed system. Existing seed/body decoders, primitive-edge label algebras and downstream separating-record witnesses have different typed domains and cannot be reused as q_{P_3} without an additional source theorem.

The normalization in Eq. (36) is not an independent physical parent-law coefficient if a complete typed role differential is already source-owned. For a surjection $R_{P_3} : \mathcal{P}_{P_3}^1 \rightarrow E_{P_3}^{\text{lbl}}$, define

$$H_R = R_{P_3}G^{-1}R_{P_3}^* > 0, \quad q_{P_3} = H_R^{-1/2}R_{P_3}. \quad (38)$$

The unique positive square root gives $q_{P_3}G^{-1}q_{P_3}^* = I_7$ and $\ker q_{P_3} = \ker R_{P_3}$.

The kernel and no-bypass clauses can be reduced further. Let $\{L_\alpha\}_{\alpha \in \mathfrak{S}_{P_3}^1}$ be the complete declared family of degree-one P3 operator and terminal jets on $\mathcal{P}_{P_3}^1$, and set

$$L_\oplus = \bigoplus_\alpha L_\alpha, \quad N_{P_3}^1 = \ker L_\oplus, \quad E_{P_3}^{\text{phys}} = \mathcal{P}_{P_3}^1 / N_{P_3}^1. \quad (39)$$

Proposition 2 (Canonical complete-first-jet quotient). *With the minimum-lift metric induced by G , the canonical quotient map $q_{\text{null}} : \mathcal{P}_{P_3}^1 \rightarrow E_{P_3}^{\text{phys}}$ is surjective, has kernel $N_{P_3}^1$, and every L_α factors uniquely through it. Its horizontal metric adjoint is the minimum-norm isometric section. Any other coisometric quotient with the same kernel differs only by an isometry of $E_{P_3}^{\text{phys}}$, restricted to a typing-preserving isometry when the role labels are retained. Moreover,*

$$\dim E_{P_3}^{\text{phys}} = \text{rank } L_\oplus. \quad (40)$$

Consequently a complete rank-seven first-jet family supplies the carrier used in Proposition 1 without an additional role-decoder, normalization or degree-one-bypass choice.

Proof. The first isomorphism theorem gives the kernel, surjectivity and Eq. (40). Since $N_{P_3}^1 \subseteq \ker L_\alpha$ for every α , each map is constant on quotient fibres and therefore factors uniquely. Positivity of G gives the orthogonal splitting $\mathcal{P}_{P_3}^1 = N_{P_3}^1 \oplus (N_{P_3}^1)^\perp$; restriction of the quotient map to the horizontal summand defines the minimum-lift metric and the isometric section. Two coisometries with the same kernel restrict to isometries from this same horizontal summand, so their relative map is an isometry. QED. $\square$

If a proposed additional degree-one row does not annihilate $N_{P_3}^1$, then it does not constitute an independent bypass parameter of the same complete quotient: it strictly refines the joint null and proves that the declared family was incomplete. Thus the remaining upstream premise is only that the physical base-seed law supplies the complete family in Eq. (39) with rank seven. The current seed-presentation candidate has the abstract ingredients from which such a family could be formed, but the present seed grammar proves neither its completeness nor its rank in the physically realized packet.

This upstream qualification does not reopen the three downstream levels already assembled in the enriched MVP: common-source terminal factorization, the occupied P3 algebra-record-GNS packet, and the conditional terminal recoveries remain intact under their declared assumptions. The unresolved statement is only that the unrestricted physical base-seed parent must select that joint packet.

The rank premise has an exact finite source certificate. Let $K_{\text{src}} > 0$ be the frozen source Hessian on the true-null-quotiented first-jet module and let $L_{\oplus}$ have seven typed rows. Define

$$G_{P3}^{(1)} = L_{\oplus} K_{\text{src}}^{-1} L_{\oplus}^*. \quad (41)$$

Proposition 3 (Rank-seven source certificate and current-parent no-go). *The following statements are equivalent:*

$$\text{rank } L_{\oplus} = 7 \iff G_{P3}^{(1)} > 0 \iff \lambda_{\min}(G_{P3}^{(1)}) > 0 \iff \det G_{P3}^{(1)} > 0. \quad (42)$$

Strict source stability, nonzero activation of all seven typed rows, equal primitive row costs and a common action unit do not imply these conditions. Consequently the conditional three-level closure cannot be inverted to prove rank seven.

Proof. Equation (41) is the Gram matrix of $L_{\oplus} K_{\text{src}}^{-1/2}$, and $K_{\text{src}}^{-1/2}$ is invertible. Therefore its rank equals $\text{rank } L_{\oplus}$; the remaining equivalences are the finite-dimensional full-rank criteria for a positive semidefinite 7×7 matrix. For non-entailment, take the same source space, $K_{\text{src}} = I_7$, load and stationary point. Let $L_7 = I_7$, whereas L_6 has rows $e_1, \dots, e_6$ and $(e_1 + \dots + e_6)/\sqrt{6}$. Both maps have seven nonzero unit rows and total row-Gram trace seven, but $\text{rank } L_7 = 7$ and $\text{rank } L_6 = 6$. Hence all the listed upstream properties hold in both controls while Eq. (42) holds only in the first. QED. $\square$

The PS1 triangular dependency action already supplies the untyped rank on its own codomain. Let π_{dep} delete the root residual and define

$$D_{\text{dep}} = \pi_{\text{dep}} D_{\Sigma} : \bigoplus_{i=0}^7 E_i \longrightarrow Z_{\text{dep}} := \bigoplus_{i=1}^7 E_i. \quad (43)$$

Proposition 4 (Onto dependency response and typed-extraction reduction). *The map D_{dep} is surjective and*

$$D_{\text{dep}} H_{\text{src}}^{-1} D_{\text{dep}}^* = \mathcal{A}_*^{-1} \pi_{\text{dep}} K_{\Sigma}^{-1} \pi_{\text{dep}}^* > 0. \quad (44)$$

If a source-owned typed extraction $T_{P3} : Z_{\text{dep}} \rightarrow E_{P3}^{\text{lbl}}$ realizes the complete P3 first-jet family by $L_{\oplus} = T_{P3} D_{\text{dep}}$, then

$$\text{rank } L_{\oplus} = \text{rank } T_{P3}. \quad (45)$$

Thus rank seven follows exactly when this complete typed extraction is onto. The present PS1 action does not select that extraction.

Proof. The restriction of D_{dep} to $\bigoplus_{i=1}^7 E_i$ is block lower triangular with identity diagonal, so it is invertible. Inverting $H_{\text{src}} = \mathcal{A}_* D_{\Sigma}^* K_{\Sigma} D_{\Sigma}$ and substituting $D_{\text{dep}} = \pi_{\text{dep}} D_{\Sigma}$ gives Eq. (44); it is positive because it is a principal compression of $K_{\Sigma}^{-1} > 0$. Surjectivity gives $\text{im}(T_{P3} D_{\text{dep}}) = \text{im } T_{P3}$, proving Eq. (45). The rank-six and rank-seven extraction controls in Proposition 3 share the same dependency action, so that action cannot select between them. QED. $\square$

The seven dependency blocks $S1, \dots, S7$ are not the seven generally non-scalar P3 ownership roles $\{B, O, R, U, Q, M, G\}$. Consequently this result does not authorize a label-based identification. It shows instead that raw seven-direction source response is already available: the remaining unrestricted physical datum is one sourced, complete and onto P3-typed extraction. No new carrier, decoder gain or terminal normalization is required, and downstream terminal maps may not be used to choose it.

The common represented trace does not by itself construct that extraction. On a simple role block $\mathcal{A}_a = M_n(\mathbb{C})$, every real linear functional on $\mathcal{A}_a^{\text{sa}}$ invariant under all inner automorphisms is a multiple of the trace and therefore vanishes on the centered tangent $\mathcal{A}_{a,0}^{\text{sa}} = \{x = x^* : \omega_a(x) = 0\}$. Thus neither the trace nor the role ledger canonically chooses a noncentral role direction.

The occupied seed representation nevertheless supplies a canonical candidate when its role values are noncentral. Let $E_a : X_{\text{src}} \rightarrow \mathcal{A}_a^{\text{sa}}$ be the source-owned represented role evaluation and define

$$h_a = E_a(x_*) - \omega_a(E_a(x_*))I, \quad \ell_a(\delta x) = \frac{\text{Re } \omega_a(h_a D E_a \delta x)}{\sqrt{\omega_a(h_a^2)}}. \quad (46)$$

This is defined for $h_a \neq 0$ and is natural under simultaneous unitary conjugation of the occupied value and its derivative. Let $L_{\text{rad}} = \bigoplus_a \ell_a : X_{\text{src}} \rightarrow \mathbb{R}^7$.

Proposition 5 (Occupied-role radial extraction criterion). *There is a unique typed map $T_{P_3}^{\text{rad}} : Z_{\text{dep}} \rightarrow E_{P_3}^{\text{bl}}$ satisfying $L_{\text{rad}} = T_{P_3}^{\text{rad}} D_{\text{dep}}$ if and only if*

$$\ker D_{\text{dep}} \subseteq \ker L_{\text{rad}}. \quad (47)$$

When it exists, it is onto if and only if

$$\text{rank } L_{\text{rad}} = 7 \iff L_{\text{rad}} H_{\text{src}}^{-1} L_{\text{rad}}^* > 0. \quad (48)$$

The present source ledger does not entail either condition.

Proof. Surjectivity of D_{dep} identifies Z_{dep} with $X_{\text{src}}/\ker D_{\text{dep}}$. The universal property of this quotient gives the unique factor precisely under Eq. (47). Its image equals the image of L_{rad} , and the final equivalence is the positive Gram criterion of Proposition 3. Non-entailment is sharp: a zero h_a removes a row; two nonzero equal-cost anchors may induce the same covector and give rank six; and a covector nonzero on $\ker D_{\text{dep}}$ does not descend. QED. $\square$

This replaces an arbitrary T_{P_3} by a derivative of occupied role data whenever the physical seed representation supplies the seven anchors. It does not prove that the physical parent law supplies them, that the deleted root is invisible to all seven radial covectors, or that their Gram is positive definite. A non-radial typed differential remains possible only as separately sourced physics.

The common-unit role requires a typed correction to the all-radial candidate. If $E_U(x_*) = cI$, then its centered anchor is zero and Eq. (46) supplies no U row. If the source instead owns a differentiable positive scale/clock-unit character $u_* : X_{\text{src}} \rightarrow \mathbb{R}_+$, define

$$\kappa_U = d \log u_*|_{x_*}, \quad \ell_U^{\text{sc}} = \frac{\kappa_U}{\sqrt{\kappa_U H_{\text{src}}^{-1} \kappa_U^*}}. \quad (49)$$

Replacing the zero centered U row by ℓ_U^{sc} gives a hybrid family to which Proposition 5 applies unchanged. The common action unit normalizes a supplied nonzero unit incidence; it does not make κ_U nonzero. Hence the current source packet must still derive that scale/clock differential or another independently typed U incidence.

Proposition 6 (Current-source metric non-identifiability). *The current PS1 source ledger does not determine the hybrid seven-role Gram, even after restricting to full-rank extractions with equal row cost.*

Proof. The ledger gives the conditional common tangent J_* and the aggregate restrictions $e_{P1}J_*$ and e_OJ_* , but not the seven represented evaluation jets $E_a(x_*)$, $DE_a|_{x_*}$ or a selected $d \log u_*$. This already prevents evaluation of Eqs. (46) and (49). Non-identifiability is sharp. Whiten the fixed onto dependency response to $Q_{\text{dep}}H_{\text{src}}^{-1}Q_{\text{dep}}^* = I_7$. Set $T_0 = I_7$ and, for $0 < |c| < 1$, let T_c leave all rows fixed except $e_2^*T_c = ce_1^* + \sqrt{1-c^2}e_2^*$. Both $L_0 = T_0Q_{\text{dep}}$ and $L_c = T_cQ_{\text{dep}}$ factor through the same dependency quotient, have rank seven, and have unit diagonal Gram. However, the second Gram has off-diagonal entry c and determinant $1-c^2$, whereas the first is I_7 . Therefore the common reduct, rank and row normalization do not select the physical P3 metric. QED. $\square$

The finite missing datum is one occupied represented role-evaluation jet $\{E_a(x_*), DE_a|_{x_*}\}_{a \neq U}$ together with the source-owned unit row $d \log u_*|_{x_*}$. Supplying that packet from the physical base-seed representation would make the descent and positive-Gram tests decisive in one calculation. Choosing it from terminal behavior would instead be a fitted completion.

The common-source terminal assembly cannot be inverted to remove this missing datum. If $J_{\text{term}} = \mathcal{T}A_O$ is the stacked terminal precursor, a factorization $L_{\text{mix}} = RJ_{\text{term}}$ exists exactly when $\ker J_{\text{term}} \subseteq \ker L_{\text{mix}}$. The forward MOPR theorem gives only $\text{rank } J_{\text{term}} \leq \text{rank } A_O$ and does not establish this kernel inclusion. Terminal recovery of the source jet would therefore require an additional source-owned injectivity theorem and cannot be inferred from successful terminal limits.

The closest existing source module does not remove the gap by dimension matching. The conditional HRC role one-jet is $V_{\text{HRC}} = E_S \oplus E_S$, two copies of the three-dimensional standard S_4 representation. The six non-unit P3 entries, however, are fixed semantic ownership labels under the current type-preserving ledger; no tetrahedral action mixing them has been derived. Therefore

$$\text{Hom}_{S_4}(E_S \oplus E_S, \mathbf{1}^{\oplus 6}) = 0. \quad (50)$$

Indeed, each row of an equivariant map would be an invariant covector on a standard module, and no such nonzero covector exists. A matched $E_S \oplus E_S$ target action admits the identity intertwiner, so the obstruction is typing rather than dimension. Deriving such a target action would itself be the missing covariant role bridge and would have to preserve the distinct typed relation-support classes. Consequently the count $6 + 1 = 7$, even after adding the independent U scale row, does not construct the occupied P3 evaluation jet.

The six non-unit maps can nevertheless be reduced to one existing source object. Let $\mathcal{A}_{\text{curr}}$ be the de-duplicated primitive colimit, $j_a : \mathcal{A}_a \rightarrow \mathcal{A}_{\text{curr}}$ its canonical role maps, and let $x \mapsto \Pi_x$ be a C^1 germ of the single coherent physical unital $*$ -homomorphism. For source-declared self-adjoint anchors $z_a \in \mathcal{A}_a$, define

$$E_a(x) = \Pi_x(j_a(z_a)), \quad DE_a|_{x_*} = D\Pi|_{x_*}(j_a(z_a)). \quad (51)$$

All six non-unit evaluation jets are then restrictions of one representation one-jet. Differentiating the homomorphism and dagger identities gives

$$D\Pi(ab) = D\Pi(a)\Pi(b) + \Pi(a)D\Pi(b), \quad D\Pi(a^*) = D\Pi(a)^*, \quad D\Pi(1) = 0. \quad (52)$$

Thus the missing maps are not six independent completion choices once the physical typed presentation, anchors and cocone germ are supplied. The last identity also proves that the algebraic unit cannot generate the nonzero absolute U scale row. That row must come from a distinguished positive U -role scale generator or from the separately sourced character u_* . The remaining finite source datum is therefore one occupied cocone one-jet, its typed anchor tuple and the unit-scale differential. The current parent has not selected this packet, and cocone coherence by itself does not imply the rank-seven positive-Gram condition.

The unit-scale differential is not wholly structureless on the conditional protected relation-unit branch already isolated upstream. If the physical source occupies the commuting relation-unit square in a source-frozen chart, then the positive character may be chosen as $u_* = U_R = a_*/c_R$, and therefore

$$\boxed{\kappa_U = d \log U_R = d \log a_* - d \log c_R.} \quad (53)$$

In the common CHDF Hodge chart, where $U_R = \sqrt{2} \ell_H$, this becomes $\kappa_U = d \log \ell_H$. This is a conditional pullback identity, not an activation theorem. Equal logarithmic source responses give $\kappa_U = 0$; moreover, the finite common-unit witness fixes ratios at one packet but does not itself provide a nonzero source tangent. Thus the remaining question is physical occupation of the protected square together with nonzero activation, rather than the form of an otherwise arbitrary seventh row.

There is also an exact finite novelty certificate. Let L_6 collect the six descended non-unit rows and assume $G_6 = L_6 H_{\text{src}}^{-1} L_6^* > 0$. With $b_U = L_6 H_{\text{src}}^{-1} \kappa_U^*$, define

$$\sigma_U = \kappa_U H_{\text{src}}^{-1} \kappa_U^* - b_U^* G_6^{-1} b_U. \quad (54)$$

The block-Gram Schur-complement theorem gives

$$\text{rank} \begin{pmatrix} L_6 \\ \kappa_U \end{pmatrix} = 7 \iff \sigma_U > 0. \quad (55)$$

Thus even a nonzero unit differential is insufficient when it lies in the source-metric row span of the six previous roles. Physical activation needs protected-square occupation, quotient descent of κ_U , and the positive certificate (54).

The unchanged source inventory cannot supply that activation. The derived source arrows form a finite S_4 relabeling groupoid with zero continuous tangent dimension, while the occupied CHDF seed orbit is locally isolated with zero on-shell seed-moduli tangent. Therefore

$$T_{x_*} X_{\text{phys}}^{\text{curr}} = 0 \implies d \log U_R|_{T_{x_*} X_{\text{phys}}^{\text{curr}}} = 0. \quad (56)$$

The available off-shell seed-body response germ is not a selected family of physical source arrows and cannot replace this missing tangent.

The adopted MCSCL forward completion provides a conditional reopening. Its common calibration tangent c_U satisfies

$$D \log \ell_H(c_U) = 1, \quad D \log \mathcal{A}_*(c_U) = 2, \quad \text{hence} \quad \kappa_U(c_U) = 1 \quad (57)$$

on the common CHDF/PRUB chart. Calibration-basiness of the six raw rows is not an additional physical premise. For an arbitrary descended six-row family define its measured common-scale loading and relative part by

$$v_U := L_6 c_U, \quad L_6^{\text{rel}} := L_6 - v_U \kappa_U. \quad (58)$$

Then $L_6^{\text{rel}} c_U = 0$ and

$$\begin{pmatrix} L_6^{\text{rel}} \\ \kappa_U \end{pmatrix} = \begin{pmatrix} I_6 & -v_U \\ 0 & 1 \end{pmatrix} \begin{pmatrix} L_6 \\ \kappa_U \end{pmatrix}. \quad (59)$$

The block-triangular row transformation is source-determined, invertible and has determinant one. It preserves the complete joint kernel and rank, while the source-metric Gram changes only by congruence. Since the relative rows annihilate c_U but $\kappa_U(c_U) = 1$,

$$\text{rank} \begin{pmatrix} L_6 \\ \kappa_U \end{pmatrix} = 7 \iff \text{rank} L_6^{\text{rel}} = 6 \iff L_6^{\text{rel}} H_{\text{src}}^{-1} (L_6^{\text{rel}})^* > 0. \quad (60)$$

Thus the common scale is counted exactly once without discarding it. This removes raw calibration-basiness as a completion assumption, but does not manufacture a missing role direction. The unrestricted physical parent must still own the MCSCL character, the protected relation-unit square, the six occupied evaluation jets, and their positive rank-six relative Gram.

The onto dependency response already supplies the underlying relative carrier. Let $D_{\text{dep}} : X_{\text{src}} \rightarrow Z_{\text{dep}}$ be the surjective seven-dimensional map above and assume quotient descent

$$\kappa_U = \bar{\kappa}_U D_{\text{dep}}, \quad \bar{\kappa}_U(D_{\text{dep}} c_U) = 1. \quad (61)$$

With $z_U = D_{\text{dep}} c_U$, define

$$E_{\text{dep}}^{\text{rel}} := \ker \bar{\kappa}_U, \quad P_{\text{rel}} := I - z_U \bar{\kappa}_U. \quad (62)$$

Direct multiplication gives

$$P_{\text{rel}}^2 = P_{\text{rel}}, \quad \text{im } P_{\text{rel}} = E_{\text{dep}}^{\text{rel}}, \quad \ker P_{\text{rel}} = \text{span}\{z_U\}, \quad \dim E_{\text{dep}}^{\text{rel}} = 6. \quad (63)$$

Restriction of the positive dependency metric to this kernel is positive definite. Hence no additional six-dimensional carrier or metric is needed. What remains unselected is the occupied typed isomorphism $T_{\text{rel}} : E_{\text{dep}}^{\text{rel}} \rightarrow E_{P_3}^{\neq U} = \text{span}\{B, O, R, Q, M, G\}$. Orthogonal frames of the same relative carrier have identical upstream status but different semantic assignments, so positivity alone cannot choose T_{rel} .

The existing typed support ledger reduces, but does not remove, this ambiguity. On the non-unit role set retain the named supports

$$H_{P_4} = \{B, M, G\}, \quad H_{OI} = \{O, Q\}, \quad H_{PM} = \{R, M, G\}. \quad (64)$$

Here H_{P_4} is the non-unit, non- I restriction of the maximal P_4 support, while H_{OI} and H_{PM} are the observer-instrument and photon-matter relation supports. The incidence fingerprints with respect to this ordered support family are

$$\begin{array}{c|cccccc} a & B & O & R & Q & M & G \\ \hline \chi(a) & 100 & 010 & 001 & 010 & 101 & 101. \end{array} \quad (65)$$

Therefore the permutation stabilizer of every named support is exactly

$$\text{Aut}(\mathcal{H}_{P_3}^{\text{rel}}) \simeq S_{\{O, Q\}} \times S_{\{M, G\}}, \quad |\text{Aut}(\mathcal{H}_{P_3}^{\text{rel}})| = 4. \quad (66)$$

Indeed, B and R have unique fingerprints; the only repeated columns are the O/Q and M/G pairs, and either pair swap preserves all three supports. Consequently, if these supports descend to six occupied atomic rays of $E_{\text{dep}}^{\text{rel}}$, support preservation reduces the semantic label freedom to two binary choices. A source-owned distinction on each unresolved pair is then necessary and sufficient to make the permutation stabilizer trivial; one common source object may supply both distinctions. This is not yet a construction of T_{rel} : the current PS1 action proves neither the atomic support descent nor those two oriented separators. It is a finite sharpening of the missing datum, not permission to select the pair orientations from terminal behavior.

This remainder has an exact quotient formulation. Put

$$q_{\text{rel}} := P_{\text{rel}} D_{\text{dep}} : X_{\text{src}} \longrightarrow E_{\text{dep}}^{\text{rel}}. \quad (67)$$

Let $\Xi_{\text{src}} : X_{\text{src}} \rightarrow V_{\text{sep}}$ be one source-owned descriptor. Since q_{rel} is onto, there is a unique descended map $\bar{\Xi} : E_{\text{dep}}^{\text{rel}} \rightarrow V_{\text{sep}}$ satisfying $\Xi_{\text{src}} = \bar{\Xi} q_{\text{rel}}$ exactly when

$$\ker q_{\text{rel}} \subseteq \ker \Xi_{\text{src}}. \quad (68)$$

Proposition 7 (Relative pair-selector criterion). *Assume that the supports in Eq. (64) descend to occupied atomic rays with source-owned normalized representatives $r_a \in E_{\text{dep}}^{\text{rel}}$. The subgroup that preserves both those supports and the values $\xi_a := \Xi(r_a)$ is trivial if and only if*

$$\boxed{\xi_O \neq \xi_Q \quad \text{and} \quad \xi_M \neq \xi_G.} \quad (69)$$

Thus a single common source descriptor can remove both residual binary ambiguities without a new carrier or role-dependent gain.

Proof. Equation (66) shows that the support stabilizer is generated by the independent swaps $(O \ Q)$ and $(M \ G)$. The first preserves the descriptor exactly when $\xi_O = \xi_Q$, and the second exactly when $\xi_M = \xi_G$. Their common stabilizer is therefore trivial precisely under Eq. (69). The descent statement is the universal property of the onto quotient q_{rel} . $\square$

The irreducible observer–source incidence and post-body variational ownership have the right shape for a two-component candidate: one could distinguish O/Q , the other M/G . Existing rank and terminal-variation theorems do not, however, prove that they form one source-owned descriptor, satisfy Eq. (68), or obey Eq. (69). Terminal meanings cannot supply those parent-side facts circularly. The finite certificate checks the order-four, one-pair order-two, both-pair trivial and nonfactorizing-null controls; it does not assert physical base–seed occupation of Ξ_{src} .

There is a further exact restriction on the natural candidate. Let Δ_{rec} be the present commutative special dagger–Frobenius copy, S the exchange of its two tensor legs and $P_- = (I - S)/2$. On its copy basis,

$$\Delta_{\text{rec}}(e_k) = e_k \otimes e_k, \quad S\Delta_{\text{rec}} = \Delta_{\text{rec}}, \quad P_- \Delta_{\text{rec}} = 0. \quad (70)$$

Proposition 8 (Current-candidate no-go and ordered-port target). *Relative to the source objects presently constructed, irreducible observer–source incidence, post-body matter/geometry variational typing and the commutative Frobenius copy do not define a noncircular Ξ_{src} satisfying Proposition 7. A sufficient finite replacement is one source-owned ordered typed port incidence*

$$\Pi_{\text{port}} : X_{\text{src}} \longrightarrow VOQ \oplus VMG, \quad \Pi_{\text{port}} = \bar{\Pi}_{\text{port}} q_{\text{rel}}, \quad (71)$$

whose descended values distinguish both O/Q and M/G .

Proof. The observer–source construction supplies a transverse image and rank, but turning it into an O/Q value requires an additional source-owned codomain functional or port assignment. Matter/geometry variational ownership is typed only after the semantic P3 roles have been identified; pulling it back to $E_{\text{dep}}^{\text{rel}}$ through the unknown T_{rel} would use the desired identification in its own construction. Equation (70) shows that the current copy has no exchange-odd component. The current objects therefore coexist with both residual pair swaps and do not imply Eq. (69). Conversely, Eq. (71) obeys the quotient descent condition by construction, and Proposition 7 removes the complete residual stabilizer when its two pairwise inequalities hold. $\square$

The previously derived exchange-odd charged-wall operator $D_{12} = Y_A \otimes I - I \otimes Y_A$ is a type witness for such orientation, not this selector. It acts on a conditional charged two-leg realization, the current parent inventory does not own its required E_{60} carrier, and no factorization through q_{rel} has been established. Proposition 8 is therefore a no-go for the current source inventory, not a theorem that an ordered port incidence cannot exist. Its physical base–seed occupation remains open.

The existing universal-seed orientation is only one binary sign. This gives an additional finite no-go.

Proposition 9 (Single-seed-sign no-go). *Let $G_{\text{pair}} = S_{\{O,Q\}} \times S_{\{M,G\}} \simeq \mathbb{Z}_2 \times \mathbb{Z}_2$ be the residual support stabilizer. No equivariant action of G_{pair} on one binary seed-orientation line is faithful. Thus the existing scalar seed sign alone cannot orient both O/Q and M/G .*

Proof. Such an action is a character $\chi : G_{\text{pair}} \rightarrow \{+1, -1\} \simeq \mathbb{Z}_2$. Its image has order at most two, whereas $|G_{\text{pair}}| = 4$. The first isomorphism theorem therefore gives $|\ker \chi| \geq 2$. At least one nontrivial pair swap remains invisible. $\square$

This result does not multiply the universal morphology seed. One structured seed or joint base-seed object may carry the faithful bidegree (χ_{OQ}, χ_{MG}) with trivial common kernel. What is excluded is only identifying the single global causal/time orientation bit with the complete relative-role selector. The two typed characters, or an equivalent ordered port incidence, still require a source derivation and quotient descent.

The required two-character object nevertheless has a canonical finite representation once its source ownership is supplied. Let $K_{\text{rel}} > 0$ be the relative metric and let J_{OQ}, J_{MG} be commuting K_{rel} -isometric involutions. Assume each has a source-oriented one-dimensional negative eigenspace, spanned by unit vectors c_{OQ} and c_{MG} , with each anchor even under the other involution. Define

$$\bar{\Pi}_{\text{port}}(v) = (K_{\text{rel}}(c_{OQ}, v), K_{\text{rel}}(c_{MG}, v)). \quad (72)$$

Proposition 10 (Ordered-port representation). *Suppose J_{OQ} exchanges r_O, r_Q and fixes r_M, r_G , whereas J_{MG} exchanges r_M, r_G and fixes r_O, r_Q . If*

$$K_{\text{rel}}(c_{OQ}, r_O) \neq 0, \quad K_{\text{rel}}(c_{MG}, r_M) \neq 0, \quad (73)$$

then Eq. (72) separates both residual pairs and has trivial support stabilizer. If the raw involutions and anchors preserve $\ker q_{\text{rel}}$, its pullback through q_{rel} is the source descriptor required by Proposition 7.

Proof. Isometry and oddness give $K_{\text{rel}}(c_{OQ}, r_Q) = K_{\text{rel}}(c_{OQ}, J_{OQ}r_O) = -K_{\text{rel}}(c_{OQ}, r_O)$, and the first condition in Eq. (73) makes these values distinct. The same argument applies to M/G . Proposition 7 then gives the trivial stabilizer, while preservation of $\ker q_{\text{rel}}$ is precisely the quotient-descent condition. $\square$

The conditional oriented observer-source double cover already present in the hub is a candidate for the first odd line, and the once-counted matter/geometry variational split motivates the second. The current physical source ledger proves neither joint occupation of those two oriented lines nor their preservation of $\ker q_{\text{rel}}$. Proposition 10 is therefore a finite sufficient realization theorem, not unrestricted parent-law realization.

The two unresolved pairs nevertheless share one variational structure in the standard leaf. A normalized observer effect evaluates a quantum state, whereas a metric variation evaluates matter stress:

$$\mathfrak{e}_{OQ}(E, \rho) = \text{Tr}(\rho E), \quad \mathfrak{e}_{MG}(h, T) = -\frac{1}{2} \int d^4x \sqrt{-g} T^{\mu\nu} h_{\mu\nu}. \quad (74)$$

These terminal identities do not derive a parent carrier. They do require every differentiable completion that recovers them to retain two typed source/response dualities. Let $i \in \{OQ, MG\}$, let $W_i \subset E_{\text{dep}}^{\text{rel}}$ be the protected pair subspace, let $h_i > 0$ be its inherited metric, let $\bar{\eta}_i \in (E_{\text{dep}}^{\text{rel}})^*$ be a nonzero covector supported on W_i , and admit one conjugate response $v_i \in W_i$. Assume a source-owned variance representation maps the source and response legs to the two occupied role rays of the pair. Consider

$$\Gamma_{\text{port}}^{\text{CTC}} = \sum_i [\Psi_i(v_i) + \Psi_i^*(\bar{\eta}_i) - \bar{\eta}_i(v_i)] = \frac{1}{2} \sum_i \|v_i - \bar{\eta}_i^\# \|_{h_i}^2, \quad \Psi_i(v) = \frac{1}{2} \|v\|_{h_i}^2. \quad (75)$$

Proposition 11 (Minimal bicotangent port completion and selection boundary). *In the local positive quadratic one-copy class with inherited metrics and one common action unit, Eq. (75) is the unique completion that has the exact conjugate graphs $v_i = \bar{\eta}_i^\sharp$, zero minimized source remainder and no pair-dependent gain. Its two source/response exchanges are commuting isometric involutions with one-dimensional odd contrast lines. If the source pullbacks $\eta_i^{\text{src}} = \bar{\eta}_i q_{\text{rel}}$ are nonzero and the declared variance representation reaches the two role rays, those lines give the descriptor of Proposition 10.*

The frozen current base–seed/body reduct does not entail this completion or physical occupation of both odd lines.

Proof. Fenchel–Young gives each term in Eq. (75) as a nonnegative square, with equality exactly at $v_i = \bar{\eta}_i^\sharp$. In a metric-normalized one-mode chart its joint Hessian is

$$H_i = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad (76)$$

whose response Schur complement is zero. For a general normalized quadratic candidate with coefficients (a_i, b_i, c_i) , exact conjugacy gives $c_i = a_i$, zero Schur remainder gives $b_i = c_i^2/a_i = a_i$, and the inherited vertical curvature gives $a_i = 1$. Hence the completion is unique in the declared class. Leg exchange is an isometry of H_i ; the two exchanges act on disjoint blocks and commute, and their odd lines are generated by the two source–response differences. Source basicness is precisely the quotient descent condition, so Proposition 10 applies.

For non-entailment, the hub’s cotangent-role control contains two models with the same base, universal seed, stabilized body, protected covector, metric and action unit: one omits the conjugate response and one admits it. Their response ranks differ. Independently, an endpoint-directed source and an oriented double-cover presentation can share the same local functional, and the endpoint $\mathbb{Z}_2^{OS}$ is not the P3 O/Q swap without a typed intertwiner. Products of these controls have identical frozen reducts but different ordered-port content. The current reduct therefore cannot select the bicotangent packet. This is an unrestricted parent-law non-entailment, not a defect of the sufficient completion. $\square$

Thus standard Born and Hilbert-stress recovery select Eq. (75) as the weakest source-preserving completion in the declared local class. They do not prove that the earlier physical base–seed reduct admits and occupies both quotient-basic covectors or owns their variance representation on the P3 rays. No further port coefficient or functional-shape search is licensed; the open question is cotangent-role and typed-port admission by the enriched parent.

The hub’s exterior and record results reduce the arbitrariness of that handoff. Let

$$\mathcal{F}_{P3} = \Lambda^\bullet E_{P3} \otimes \mathcal{R}_2, \quad \dim \mathcal{R}_2 = 2, \quad (77)$$

where $\mathcal{R}_2$ is the already selected two-leg dagger–Frobenius record carrier. On $\Lambda^\bullet E_{P3}$ define

$$\mathbf{C}_{\text{ext}}(v) = \varepsilon(v) + \iota_{K_*(v, \cdot)}. \quad (78)$$

The canonical exterior identities give $\mathbf{C}_{\text{ext}}(v)^2 = K_*(v, v)I$, and tensoring with $I_{\mathcal{R}_2}$ leaves this law unchanged. The exterior carrier has dimension $2^7 = 128$ with equal central sectors of dimension 64; the record leg doubles them to $(128, 128)$. Since each simple $M_8(\mathbb{C})$ module has dimension eight, the resulting multiplicities are $(16, 16)$ and the normalized trace weights are $(1/2, 1/2)$.

This is a functorial, coefficient-free realization of the premise of Proposition 1; it does not identify the seven Paper-IV dependency hyperedges with the seven P3 roles. Its physical premise is

instead joint source ownership and occupation of the already typed E_{P3} role quotient, its positive metric and the two-leg record carrier. Those ingredients exist inside the declared source-irredundant enriched MVP. More precisely, if $r_* = \Delta_\tau e_*$ is its normalized occupied Frobenius record state, then

$$J_{P3,R}\psi = \psi \otimes r_*, \quad J_{P3,R}^\dagger J_{P3,R} = I, \quad (79)$$

and $(C_{\text{ext}}(v) \otimes I)J_{P3,R} = J_{P3,R}C_{\text{ext}}(v)$. Thus the common-source lift cannot annihilate occupied P3 support and adds no gain: joint assembly is closed inside that declared MVP carrier class. This conclusion is not a consequence of marginal occupation alone. On $\mathbb{C}^2$, the orthogonal projectors $P = |0\rangle\langle 0|$ and $R = |1\rangle\langle 1|$ both act nontrivially on $(|0\rangle + |1\rangle)/\sqrt{2}$, while $PR = 0$. Consequently the unrestricted base-seed reduct still does not imply that its physical parent law realizes and occupies the common enriched MVP carrier, but no additional internal joint-occupation gate remains after that carrier is fixed.

The same packet removes the variance-representation premise left explicit in Proposition 11. Define

$$\Phi_{P3}(v) = C_{\text{ext}}(v) \otimes I_{\mathcal{R}_2}, \quad \omega_{P3,R}(A) = \frac{1}{256} \text{Tr}(A). \quad (80)$$

Proposition 12 (Canonical port variance and operational occupation). *On the occupied typed P3 exterior-record packet,*

$$\omega_{P3,R}(\Phi_{P3}(v)) = 0, \quad \text{Re } \omega_{P3,R}(\Phi_{P3}(u)^* \Phi_{P3}(v)) = K_*(u, v). \quad (81)$$

Hence Φ_{P3} is a faithful metric-isometric variance representation and its restrictions to the O/Q and M/G pair subspaces supply the representation assumed in Proposition 11 without a new gain or carrier.

Assume in addition complete degree-one factorization through the typed P3 quotient. A normalized POVM outcome family induces a nonzero O/Q first jet. On every test region where the renormalized matter stress is a nonzero distribution, metric variation induces a nonzero M/G first jet. Their quotient covectors are basic, and the positive Riesz maps give unique nonzero bicotangent responses. Thus no further internal port-occupation bit is required on such support.

Proof. The operators exchange the even and odd exterior grades, so their normalized trace is zero. The exterior Clifford relation gives

$$\Phi_{P3}(u)\Phi_{P3}(v) + \Phi_{P3}(v)\Phi_{P3}(u) = 2K_*(u, v)I. \quad (82)$$

The operators are self-adjoint; normalized-trace cyclicity proves Eq. (81). Setting $u = v$ gives $\|\Phi_{P3}(v)\|_{2,\omega}^2 = K_*(v, v)$, so positivity of K_* proves injectivity and nonzero images for all typed role rays.

For the state-dependent claim, POVM normalization gives $\sum_a \text{Tr}(\rho_O E_a) = 1$, so at least one typed effect-source derivative is nonzero. A nonzero stress distribution is exactly a nonzero linear functional on compactly supported metric test variations, hence some h has $-\frac{1}{2} \int \sqrt{-g} T^{\mu\nu} h_{\mu\nu} \neq 0$. Complete typed factorization descends both jets to nonzero quotient covectors; their source pullbacks annihilate the quotient kernel. Since a positive metric has an invertible Riesz map, both conjugate responses are nonzero. QED. $\square$

The nonzero-stress hypothesis is necessary. An identically zero renormalized stress on a region gives a zero M/G first jet even though the abstract M and G role rays and their faithful representation remain present. Proposition 12 therefore closes an internal representation/activation issue

of the already occupied enriched MVP; it does not prove that the unrestricted physical base–seed parent selects the typed quotient, the Clifford–record packet or a nonzero-stress observer support.

The same construction fixes the representation class rather than merely its dimension. Since $\text{Cl}_7(\mathbb{C}) \simeq M_8(\mathbb{C}) \oplus M_8(\mathbb{C})$ has dimension 128, its left regular module contains eight copies of each simple eight-dimensional module. These are exactly the multiplicities of $\Lambda^\bullet E_{P_3}$. The exterior vacuum has full Clifford-word orbit rank 128; after the two central record projectors are included, the joint orbit rank is 256. Hence the exterior–record packet is a faithful cyclic regular representation of its P3–record algebra. Its represented normalized trace fixes the equal Clifford-sector weights, so no separate cyclicity, GNS carrier or central-weight choice remains on this subpacket.

B Complete-incidence characterization

This does not establish minimal-GNS faithfulness or differential zero cokernel for the full S_0 – S_7 source signature. The positive connected extension $\Gamma_{\text{ext}}(z, w) = \Gamma_{P_3, R}(z) + h\|w - bz\|^2/2$ has a generically nonzero raw coordinate while exact elimination gives $\text{Schur}_w H_{\text{ext}} = H_{P_3, R}$. It therefore disproves raw coordinate exhaustivity. It is not, however, a distinct physical-source completion under the adopted source ontology: in the defect coordinate $u = w - bz$ it is the centered positive auxiliary $\Gamma_{P_3, R}(z) + h\|u\|^2/2$ with $u_* = 0$, zero independent load and Schur-neutral elimination. Physical exhaustivity modulo presentation auxiliaries is instead reduced to uniqueness of the occupied stable branch and zero cokernel of the complete typed source incidence. Those two premises remain open for the unrestricted physical base–seed parent. On the declared source-frozen finite MVP action, strict positivity already supplies the unique stationary branch, so its sole remaining internal exhaustivity test is the typed zero-cokernel condition. Physical base–seed realization and occupation of that MVP action remain an upstream physical assumption.

The preceding construction can nevertheless be compressed into one exact source certificate. Let $P_{P_3}^1$ be the physical degree-one source quotient, with its declared seven typed roles, and let

$$\mathcal{C}_{P_3}^{\text{phys}} : P_{P_3}^1 \longrightarrow \mathcal{A}_{\text{body}}^{\text{sa}} \quad (83)$$

be a source-owned incidence into the self-adjoint part of the occupied body algebra. Call Eq. (83) *complete* when (i) its quotient rank is seven and its kernel is exactly the complete P3 physical null, with no terminal-visible bypass; (ii) after polar normalization it obeys

$$\mathcal{C}(v)^2 = K_*(v, v)I; \quad (84)$$

and (iii) the canonical exterior extension and the two-leg dagger–Frobenius record algebra used in Eq. (79) are jointly occupied on the same source carrier by a cyclic vector with full algebra orbit (rank 256 in the present finite realization). Here *minimal* means generated by the physical degree-one source quotient with no additional P3 bypass; it does not mean smallest Hilbert-space dimension. The declared recovery class uses the canonical cyclic regular exterior–record realization.

Theorem 7 (Minimal physical P3 certificate). *Within the finite source-irredundant Clifford–recovery class, the physical parent contains the minimal faithful typed P3 Clifford–record subpacket, uniquely up to typed unitary equivalence, if and only if it owns a complete incidence (83).*

Proof. For sufficiency, the exact-null condition descends the incidence faithfully to the seven-dimensional typed quotient. Polar normalization fixes the positive metric and Eq. (84); polarization then gives the Clifford anticommutator. The exterior construction (78), the occupied record lift (79), and Proposition 12 provide the faithful cyclic Clifford–record representation and its normalized trace. Source irredundancy removes invisible degree-one copies, while complete-null/no-bypass

removes a second terminal-visible realization. The remaining freedom is therefore typed unitary equivalence.

Conversely, restrict a minimal faithful typed P3 Clifford–record subpacket to its grade-one self-adjoint Clifford generators and pull them back along the physical source first jet. Faithfulness gives rank seven, the physical quotient gives the exact kernel and no-bypass property, the Clifford law gives Eq. (84), and regular cyclic occupation supplies the same full exterior–record orbit. Hence the pulled-back incidence is complete. QED. $\square$

C Enriched variational ownership and ambient-extension control

The ownership question can be closed inside an explicit ordered enriched completion without fixing the Clifford map in advance. After the morphological body has been solved and frozen, vary $C \in \text{Hom}_{\mathbb{R}}(E_{P3}^{\text{phys}}, \mathcal{A}_{P3}^{\text{sa}})$ and one operator response S_{P3} . For the source-frozen occupied descriptor e_{P3} define

$$\begin{aligned} \Gamma_{P3}^{\text{enr}}[C, S_{P3}; e_{P3}] &= \frac{\mathcal{A}_*}{2} \tau \left[(S_{P3} - C(e_{P3}))^\dagger (S_{P3} - C(e_{P3})) \right] \\ &\quad + \frac{\mathcal{A}_*}{2} \int_{S(E_{P3}, K_*)} \tau \left[\Re_C(v)^\dagger \Re_C(v) \right] d\mu(v), \quad \Re_C(v) = C(v)^2 - K_*(v, v)I. \end{aligned} \quad (85)$$

This is a staged parent action: body variables are not varied in the post-body P3 stage. The two rows use the same faithful trace and action unit, and the second counts the primitive operator relation once.

Theorem 8 (Enriched-parent variational ownership). *In the declared source-irredundant enriched completion, supplemented by the complete physical P3 null, cyclic exterior–record occupation and no-bypass conditions, Eq. (85) owns and selects the minimal faithful P3 Clifford–record subpacket. Its zero locus is*

$$S_{P3} = C(e_{P3}), \quad C(u)C(v) + C(v)C(u) = 2K_*(u, v)I, \quad (86)$$

and exact elimination of S_{P3} leaves $\mathcal{V}_{\text{OS}}[C]$. Hence the handoff and source selector are not double-counted.

Proof. Both terms in Eq. (85) are nonnegative. The canonical exterior Clifford incidence together with $S_{P3} = C(e_{P3})$ reaches zero. Conversely, every global zero has vanishing graph defect and vanishing operator-square integral. The first gives the response equation in Eq. (86); faithfulness, continuity and homogeneity give $C(v)^2 = K_*(v, v)I$ for every v , and polarization gives the displayed anticommutator. Minimization over S_{P3} removes exactly the graph square and leaves Eq. (22). Theorem 7 then supplies typed-unitary uniqueness of the minimal occupied subpacket. QED. $\square$

The construction is not an arbitrary terminal repair. Its typed domain is the already fixed P3 quotient; its target is the canonical generated exterior algebra; its metric, trace and unit are inherited; body-first staging forbids readout-to-body backreaction; and it replaces rather than duplicates the fixed graph-normal handoff. Numerically, the graph term minimizes to zero for both the Clifford and commuting controls, while the reduced enriched costs remain 0 and 4/3, respectively; the zero incidence has normalized cost one.

This is a conditional source-law closure, not an unrestricted parent-realization theorem. The current reduct has no source variable typed as a complete rank-seven operator incidence. It admits the equal-Gram controls in Eq. (25), so it cannot entail the selector. Equation (22) is also not

claimed unique among all possible positive parent functionals; it is the coefficient-free representative of the declared one-copy orthogonally averaged squared-defect class. The theorem proves ownership by the explicitly declared enriched parent, not that the earlier reduct physically realizes and occupies that enrichment.

Theorem 7 is a characterization theorem, not a derivation of certificate existence from the current base-seed reduct. Nor can finite readouts establish that this selected subpacket exhausts the ambient parent. Indeed, for any visible action $\Gamma(x)$ and finite terminal stack $F(x)$, set

$$\tilde{\Gamma}(x, w) = \Gamma(x) + \frac{q}{2}(w - bx)^2 - j(w - bx), \quad \tilde{F}(x, w) = F(x), \quad q > 0, j \neq 0. \quad (87)$$

At every visible stationary point x_* , the extension has $w_* = bx_* + j/q$, preserves $F(x_*)$, and satisfies

$$\text{Schur}_w H_{\tilde{\Gamma}} = H_{\Gamma}. \quad (88)$$

Thus the finite terminal family cannot distinguish the original parent from one with an additional independently loaded source component. In the invariant defect coordinate $u = w - bx$, Eq. (87) is $\Gamma(x) + qu^2/2 - ju$: it is not an unloaded presentation auxiliary, but it is also not a second interacting realization of the same connected universe packet. The selected connected packet is exhaustive only after the independent parent-side condition $\text{coker } J_{\text{phys}} = 0$ (or an equivalent source law) is proved.

D Cross-paper operational witness handoff

The mixed ROOT-P1 operational witness and its observer-rank calculation are owned by Foundation Paper VII-A. This technical paper uses neither that witness nor its rank as a premise of the rank-seven reconstruction.

E Class-minimal support-port enriched-law certificate

The support-port reconstruction separates the data that define the proposed physical law from the conclusions generated by that law. The support projectors and common-unit character specify the admissible source domain; the cotangent, operator-square and record terms specify one staged once-counted functional; and the decoder specifies noncollapsing body realization. They are not independent terminal fits.

Proposition 13 (Variational occupation, class-relative componentwise minimality and narrow-reduct no-go). *In the finite source-irredundant one-copy class used in Theorem 2, the normalized P3 functional is nonnegative and has an explicit simultaneous zero. Consequently its global minimum is zero and every global minimizer lies on the complete packet zero locus; no separate P3 common-zero occupation assumption is required. Moreover, every clause of the proposed P3 law is required for that theorem's conclusion in the following precise sense:*

- (i) deleting the support flag leaves the B/R singleton blocks and O/Q, M/G planes unidentified;
- (ii) deleting either ordered cotangent port leaves the corresponding nontrivial pair swap in the typed stabilizer;
- (iii) placing z_U in the relative span lowers the source rank to six even at fixed total row cost;
- (iv) deleting the left-decoder premise permits a rank-six body realization;
- (v) deleting the operator-square term admits a commuting packet with the same normalized marginal trace Gram;
- (vi) deleting the pointed root admits a faithful full-rank nonracial orbit; and

(vii) deleting cyclicity or no bypass permits unoccupied or additional record support.

The two cotangent response legs and the graph-normal P3 response can be eliminated exactly with zero additional source stiffness, so their inclusion does not duplicate the retained operator-square law. The unchanged narrow base–seed reduct admits the first, third and fifth controls with the same declared marginal upstream data and therefore does not entail $\mathcal{L}_{P3}^{\min}$.

Proof. The shifted stabilized-body term is nonnegative on its declared basin. Each cotangent contribution is a Fenchel–Young square; the P3 graph and operator-square contributions are squared faithful-trace norms; and the record contribution is the normalized squared record defect. Choose the canonical exterior Clifford incidence, set its graph response equal to its source evaluation, use the two metric-dual ordered ports, the stabilized body and its exact left decoder, and take the occupied empty exterior word with the normalized two-leg record root. Every displayed defect then vanishes simultaneously. Hence the action floor is zero. A sum of nonnegative terms can attain that floor only when every term vanishes, proving variational occupation of the packet.

The support stabilizer is $S_{\{O,Q\}} \times S_{\{M,G\}}$. One port removes one factor and two ports remove both. The independent/dependent unit controls have ranks seven and six with equal total squared row norm. The redundant body realization has a left inverse, whereas identifying its unit column with another column lowers the rank and makes an exact left inverse impossible. Clifford generators and the commuting control have the same identity trace Gram, but only the former have a state-neutral square and vanishing mixed anticommutators. The positive noncentral root-density control has full orbit rank and equal central weights but a nonidentity Gram. The cyclic/no-bypass controls are the complete-incidence and pointed-orbit countermodels proved above. Finally, each normalized Fenchel–Young port has zero source Schur remainder, and minimizing the graph square in Eq. (85) leaves the operator-square functional exactly. These facts prove all clauses and the same-reduct non-entailment. QED. $\square$

The proposition is a class-relative minimality certificate for the stated theorem, not a claim that no richer physical parent law can realize the same packet. The superscript “min” denotes only componentwise minimality in the declared finite, source-irredundant one-copy class. Under the four-dimensional recovery-completion policy and within that class, the O/Q and M/G dualities, their de-duplicated relation supports, the common unit and body noncollapse are the weakest preimages of the already declared standard recovery targets. The state-neutral operator-square norm is the genuinely new Tau-specific cross-role clause. The pointed cyclic root is instead the minimal record-side response, within the class, to the faithful full-rank nonracial countermodel. Hence the paper no longer leaves an unspecified “complete rank-seven incidence” as a free input: it proposes the explicit law in Eq. (14), derives the packet from it, and derives occupation from global minimization rather than assuming its common zero separately. Its empirical truth and unrestricted ambient-source exhaustivity remain separate questions.

References

- Christa Fluehmann, T. L. Nguyen, Matteo Marinelli, Vlad Negnevitsky, Karan Mehta, and Jonathan P. Home. Encoding a qubit in a trapped-ion mechanical oscillator. *Nature*, 566:513–517, 2019. doi: 10.1038/s41586-019-0960-6.
- S. C. Burd, R. Srinivas, J. J. Bollinger, A. C. Wilson, D. J. Wineland, D. Leibfried, D. H. Slichter, and D. T. C. Allcock. Quantum amplification of mechanical oscillator motion. *Science*, 364(6446):1163–1165, 2019. doi: 10.1126/science.aaw2884.

Rodrigo A. Thomas, Michał Parniak, Christoffer Østfeldt, Christoffer B. Møller, Christian Bærentsen, Yeghishe Tsaturyan, Albert Schliesser, Jürgen Appel, Emil Zeuthen, and Eugene S. Polzik. Entanglement between distant macroscopic mechanical and spin systems. *Nature Physics*, 17:228–233, 2021. doi: 10.1038/s41567-020-1031-5.

Valeriy Novikov, Jun Jia, Túlio Brito Brasil, Andrea Grimaldi, Maïmouna Bocoum, Mikhail Balabas, Jörg Helge Müller, Emil Zeuthen, and Eugene Simon Polzik. Hybrid quantum network for sensing in the acoustic frequency range. *Nature*, 643:955–960, 2025. doi: 10.1038/s41586-025-09224-3.